Theoretical Minimum
Cosmology

Simon Crase (compiler)
simon@greenweaves.nz

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Abstract

These are my notes for the *Cosmology* lectures from Leonard Susskind's *Theoretical Minimum* series[1, Cosmology].

Disclaimer: I have created these notes as an aide-mémoire for my own use; if you find them useful, you are welcome, but I'd appreciate hearing from you. They are not intended as a substitute for listening to the lectures. The intellectual property for all material derived from the lectures belongs, of course, to Professor Susskind; any mistakes, however, are my own.

The notes were created using TexStudio[2], the bibliography using JabRef[3], and Feynman diagrams using *TikZ-Feynman* [4].

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1 The expanding (Newtonian) universe

Cosmology as a science dates back to the discovery of the microwave radiation from the Big Bang in the 1960s. Before that Cosmology was more like natural science–classifying oddities. Thinking of the universe as a physical system, as a system to study mathematically, with a set of physical principles and a set of equations, is relatively new.

We will study the universe as a physical system, i.e. something we can study with equations.

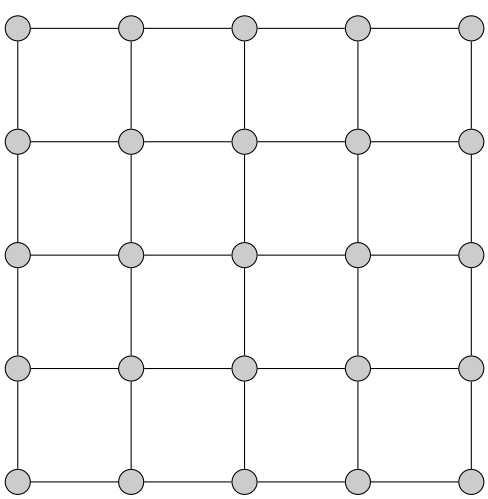
- First observation: the universe is isotropic, averaging over patches of the sky, so it should be homogenous. If not we'd have shells centered on Earth!
- There are around 10^{11} galaxies within view of Earth, which we can treat as particles. Each galaxy has roughly 10^{11} stars.
- Gravity is the only force that matters at cosmological scales: it pulls stuff together. Should stuff remain static?

Could Newton have figured out expanding universe given observations?

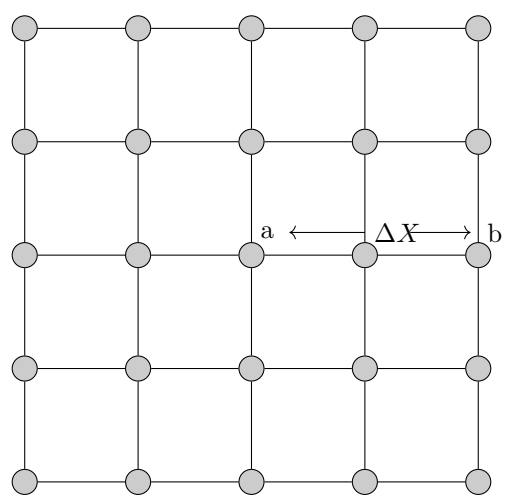
1.1 How would Newton have handled an expanding universe?

Figure 1: We start by introducing a set of coordinates.

(a) Chose grid so that galaxies are at grid points, so grid moves with galaxies. Galaxies move coherently (from observation).



(b) Define distance: $D_{ab} = a(t)\Delta x_{ab}$, where scale factor a may or may not be a constant.



We postulate a more general formula for distance between two galaxies a and b:

$$D_{ab} = a(t)\sqrt{\Delta_{ab}x^2 + \Delta_{ab}y^2 + \Delta_{ab}z^2}, \text{ where } a(t) \text{ is called the "scale factor"} \tag{1}$$

Now calculate velocity:[1]

$$\begin{aligned} V_{ab} &= \dot{a}\Delta X_{ab}, \text{ neglecting } y \text{ and } z \\ \frac{V_{ab}}{D_{ab}} &= \frac{\dot{a}}{a}, \text{ independent of choice of galaxies!} \\ &= H, \text{ Hubble's constant at a given time.} \\ V &= HD, \text{ Hubble's Law.} \end{aligned} \tag{2}$$

1.2 How much mass is in $\Delta x \Delta y \Delta z$?

Assume $\Delta x \Delta y \Delta z$ big enough that we can average over the small scale structure. How much mass is in there? Let M denote the mass within $\Delta x \Delta y \Delta z$, and $M = \nu \Delta x \Delta y \Delta z$. Let V denote the volume[1].

$$\begin{aligned} V &= a^3 \Delta x \Delta y \Delta z \\ M &= \nu \Delta x \Delta y \Delta z, \text{ constant} \\ \rho(t) &= \frac{\nu}{a(t)^3} \end{aligned} \tag{3}$$

Amount of mass in a given region of Figure 1a remains the same, as galaxies move with grid.

1.3 Newtonian derivation of Friedmann's equation

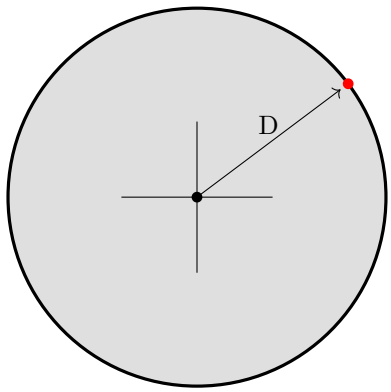
Choose grid so Newton is at rest at centre of universe. He looks out at distant galaxy, which moves in accordance with Newton's laws. He would use Newton's Theorem–Figure 2 and Theorem 1.

Theorem 1 (Newton's shell theorem)**.** *1. A spherically symmetric body affects external objects gravitationally as though all of its mass were concentrated at a point at its center.*

2. If the body is a spherically symmetric shell (i.e., a hollow ball), no net gravitational force is exerted by the shell on any object inside, regardless of the object's location within the shell.

[1] During the lecture, V is used for both velocity and volume. I will retain this usage in the notes; the reader should be careful to check which usage is meant.

Figure 2: Newton's Theorem: to determine gravitational force on red body, assuming isotropy, draw sphere and calculate force as if mass inside shell were concentrated in centre, and ignore mass outside.



$$\begin{aligned} D =& a(t)\sqrt{x^2+y^2+z^2} \\ =& a(t)R, \text{ say} \end{aligned} \tag{4}$$

R is constant, since galaxy is located at a fixed point in lattice. We want acceleration so we can use Newton's laws.

$$\begin{aligned} V =& \dot{a(t)}R \\ A =& \ddot{a(t)}R \end{aligned}$$

From Newton's Law of Universal Gravitation

$$\begin{aligned} F =& -\frac{mMG}{D^2} \\ A =& -\frac{MG}{D^2}, \text{ which should become} \\ =& \ddot{a(t)}R \end{aligned}$$

So we want

$$
\begin{aligned}
a\ddot{(t)}R =& -\frac{MG}{D^2} \\
=& -\frac{MG}{a^2R^2}, \text{ from (4)} \\
\frac{a\ddot{(t)}}{a} =& -\frac{MG}{a^3R^3}, \text{ now volume is} \\
V =& \frac{4\pi}{3}a^3R^3 \\
\frac{a\ddot{(t)}}{a} =& -\frac{4\pi}{3}G\rho, \text{ from definition of } \rho
\end{aligned} \tag{5}
$$

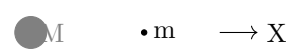
Now ρ does not depend in R, so (5) applies for every galaxy. This is true for any origin, as long as we do the transformation carefully. Everything hinges on the assumption that the universe is homogeneous.

We can see from (5) that the universe is not static unless $\rho = 0$. Let us substitute (3) in (5).

$$
\frac{a\ddot{(t)}}{a} = -\frac{4\pi}{3}\frac{G\nu}{a^3} \tag{6}
$$

This was discovered by Friedmann, in the context of General Relativity. It doesn't tell us whether universe is expanding or contracting, but tells us that acceleration is negative. But observation says that it isn't! This would have been the standard model before we found that expansion is accelerating.

Figure 3: Consider throwing rock up from Earth, along x-axis



We will write down the energy equation: total energy is $\frac{1}{2}mv^2 - \frac{mMG}{x}$. What is escape velocity if total energy is zero?

$$
\begin{aligned}
\frac{1}{2}\cancel{m}V^2 - \frac{\cancel{m}MG}{X} =& 0 \\
V^2 =& \frac{2MG}{X}
\end{aligned}
$$

Similarly in (6), total energy can by positive, negative or zero.

- If energy great enough, universe does not turn around: it expands forever.
- If energy smaller, universe eventually contracts.
- Escape velocity is the boundary between the other two cases.

In Figure 2, all the particle knows is that it is moving under influence of mass concentrated at origin, so we can treat as Figure 3.

The Total Energy is given by $\frac{1}{2}m\dot{a}^2R^2 - \frac{mMG}{aR}$. In the edge case where energy is zero:

$$\begin{aligned}
\dot{a}^2R^2 - \frac{2MG}{aR} =& 0 \\
\frac{\dot{a}^2}{a^2} - \frac{2MG\frac{4}{3}\pi}{\frac{4}{3}\pi a^3R^3} =& 0 \\
\underbrace{\left(\frac{\dot{a}}{a}\right)}_{\text{Hubble}}^2 =& \frac{8\pi}{3}G\rho
\end{aligned} \tag{7}$$

Equation (7) is known as the Friedmann equation. This universe will get slower and slower, but never collapse.

Using (3), (7) becomes:

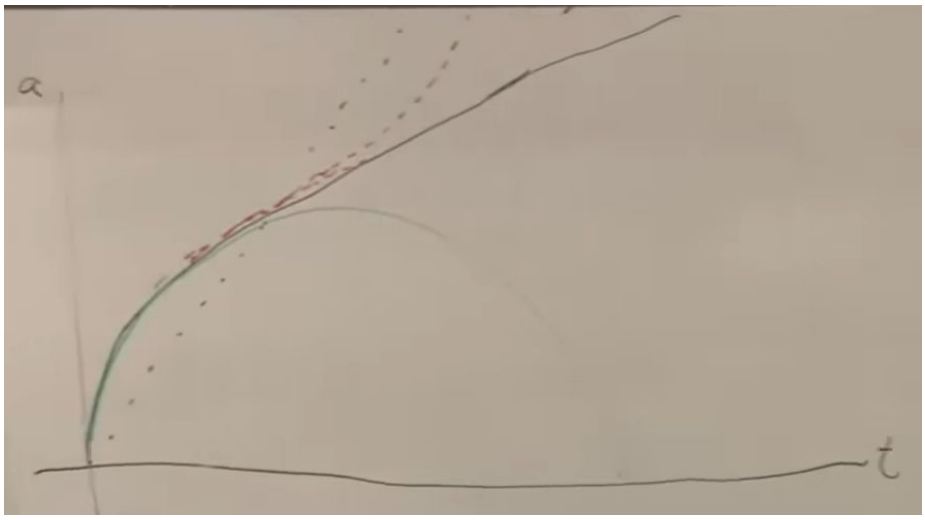
$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi\nu}{3}\frac{G}{a^3}, \text{ or, using suitable units} \tag{8}$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{1}{a^3} \tag{9}$$

The right hand side of (9) is always positive, and decreases as a increases, but slows. We will attempt a trial solution, putting $a = ct^P$.

$$\begin{aligned}
a =& ct^P \\
\dot{a} =& cPt^{P-1} \\
\frac{\dot{a}}{a} =& \frac{P}{t}. \text{ so (9) becomes:} \\
\frac{P^2}{t^2} =& \frac{1}{c^3t^{3P}}, \text{ whence} \\
3P =& 2 \\
P =& \frac{2}{3} \\
P^2 =& \frac{1}{C^3}
\end{aligned}$$

Figure 4: Scale Factor, showing slowing (case discusses), collapsing, and real (accelerating) universes.



Newton speculated about a homogeneous universe, but didn't carry out the calculation.

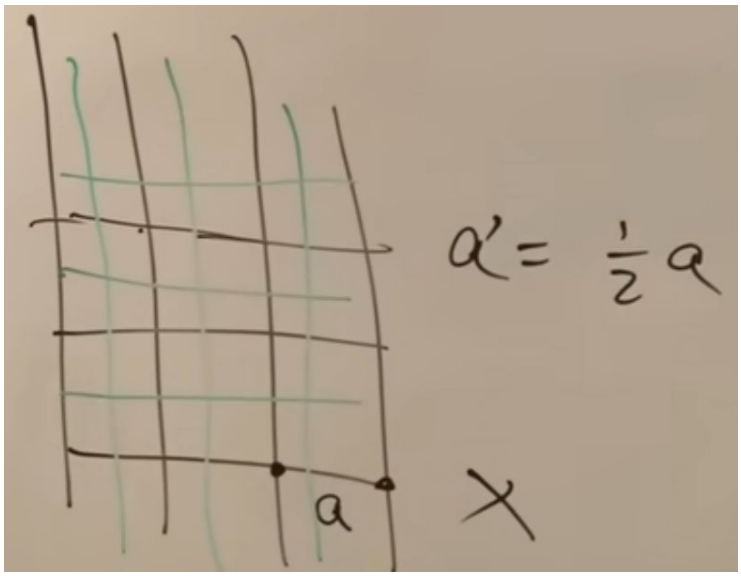
2 Matter and radiation dominated universes

Did Newton get it right? Yes, mostly. Einstein's equations deal with curved space-time, and some of the universes we will study have curved 3-space, maybe a 3-sphere. Suppose we just look at very neighbouring galaxies, where it will look flat. Then in a small region we can use Newton;s equations. So what we did in Section 1 should be OK, as long as we don't have two galaxies moving past each other at a speed comparable to that of light (But that is OK for galaxies that are a long way away).

Now photons and neutrinos are moving very fast, so we have to modify our equations to deal with this.

Treat matter in galaxies as uniform density ρ. Lay down grid as in Figure 1a. How big is spacing? Equations should not depend on spacing, but maybe the ratios matter. If a doubles, we have expansion.

Figure 5: Equations should not depend on spacing.



We could measure expansion by $\frac{\dot{a}}{a}$.

Another ambiguity is ν, which depends on a.

In Section 1 we derived (8), which reduces to:

$$(\frac{\dot{a}}{a})^2 = \frac{8\pi G\rho}{3}$$

$\frac{8\pi}{3}$ is related to the volume of a sphere. The equation assumes zero energy, or, equivalently, galaxies moving at escape velocity.

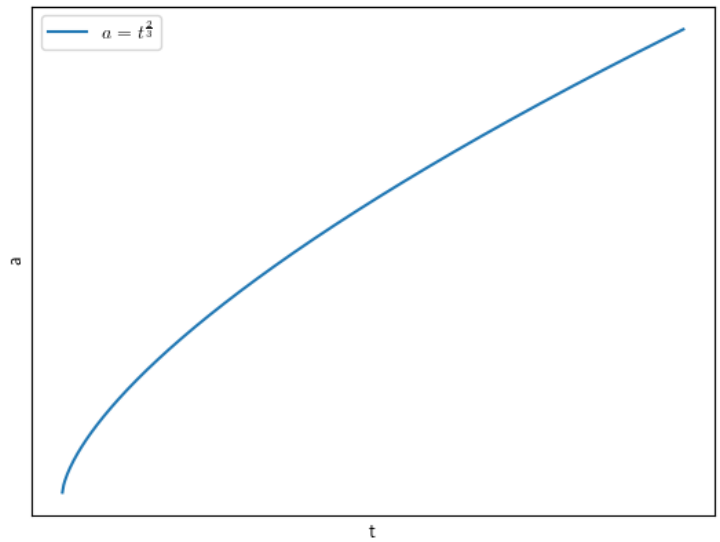
Is ν constant? If we assume, for the moment, that protons and galaxies are forever, then yes, ν is constant. Box grows, but number of particles in box remains constant. Recall (8):

$$\begin{aligned}(\frac{\dot{a}}{a})^2 &= \frac{8\pi\nu}{3}\frac{G}{a^3} \\ (\frac{\dot{a}}{a})^2 &= \frac{1}{a^3}, \text{ using suitable units} \end{aligned} \tag{10}$$

We can solve (10).

$$\begin{aligned}\dot{a} &= \frac{1}{\sqrt{a}} \\ \frac{dt}{da} &= \sqrt{a}, \pm \text{ corresponds to expansion/contraction} \\ t &= \frac{2}{3}a^{\frac{3}{2}} \\ a = t^{\frac{2}{3}}&, \text{ ignoring some constant} \end{aligned} \tag{11}$$

Figure 6: Scale parameter increases with time, but decelerates as the result of gravitational attraction

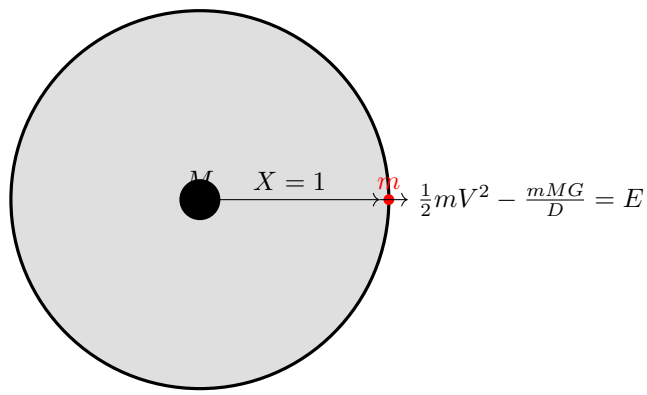


NB. Real motion of galaxies is made up of flow plus peculiar velocity. There are two directions we want to go in.

1. What if we replace matter with photons? (Early universe). Section 2.2
2. What if energy is non-zero?–Section 2.1.

2.1 Matter dominated universe

Figure 7: We compute the constant energy



$$
\begin{aligned}
V^2 - \frac{2MG}{D} =& \frac{2E}{m} \\
D =& aX \\
V =& \dot{a}X, \text{ but we have set } X = 1 \\
D =& a \\
V =& \dot{a}, \text{ so the energy equation becomes} \\
\dot{a}^2 - \frac{2MG}{a} =& C, \text{ say} \\
(\frac{\dot{a}}{a})^2 - \frac{2MG}{a^3} =& \frac{C}{a^2} \\
(\frac{\dot{a}}{a})^2 - \frac{8\pi\rho G}{3} =& \frac{C}{a^2} \\
(\frac{\dot{a}}{a})^2 =& \frac{8\pi\nu G}{3a^3} + \frac{C}{a^2}, \text{ on substituting (3)} \qquad (12)
\end{aligned}
$$

Equation (12) is the full Friedmann equation, which he derived from General Relativity; we have derived from Newton.

if $C > 0$, right hand side is strictly positive, universe continues to grow forever (For if $\dot{a} > 0$ initially, it has to pass through zero to stop expanding). If a small enough $\frac{8\pi\nu G}{3a^3}$ will dominate, and we can use (11). Once a grows enough $\frac{C}{a^2}$ dominates.

$$
\begin{aligned}
(\frac{\dot{a}}{a})^2 =& \frac{C}{a^2} \\
\frac{\dot{a}}{a} =& \frac{C}{a}, \text{ for some new } C \\
\dot{a} =& C
\end{aligned}
$$

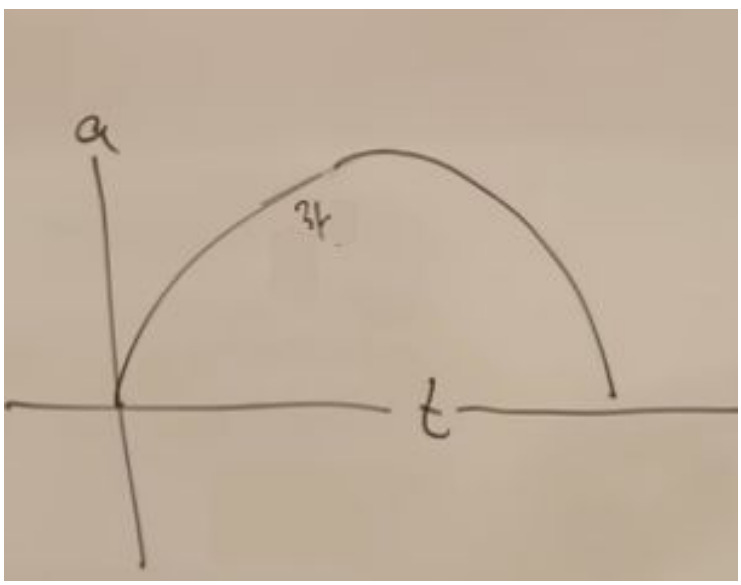
Eventually we can neglect gravity and velocity is uniform.

What if energy is negative? Equation (12) becomes:

$$
(\frac{\dot{a}}{a})^2 = \frac{8\pi\nu G}{a^3} - \frac{C}{a^2}, \text{ for some new, positive } C
$$

Now there is a point where $\dot{a} = 0$: the expansion stops, and $\dot{a}$ changes from positive branch of square root to negative.

Figure 8: Negative Energy: there is a point where $\dot{a} = 0$



The matter dominated universe was the classic cosmology until other things were discovered. There are three cases, positive energy, negative and zero. In a later lecture we will find that this is connected to the geometry of the universe: this is the main contribution of General Relativity.

2.2 Radiation dominated universe

We do have to think about relativity, but there is only one important thing: $E = mc^2$. We rewrite Equation (12) to show the connection with general relativity, and define $\kappa = C$:

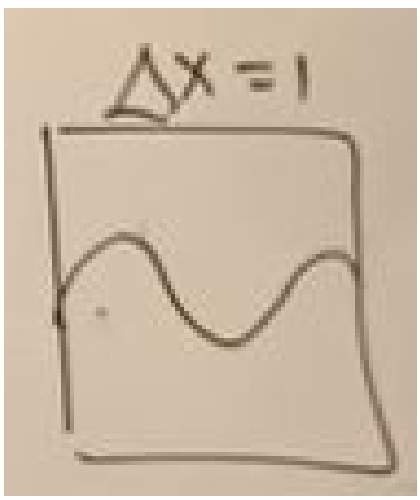
$$\underbrace{\left(\frac{\dot{a}}{a}\right)^2 - \frac{\kappa}{a^2}}_{\text{Terms involving geometry}} = \underbrace{\frac{8\pi}{3}G\rho}_{\text{Matter and energy}} \tag{13}$$

We'll replace mass density with energy density. Energy $= Mc^2 +$ Radiation (and we will choose units such that $c = 1$).

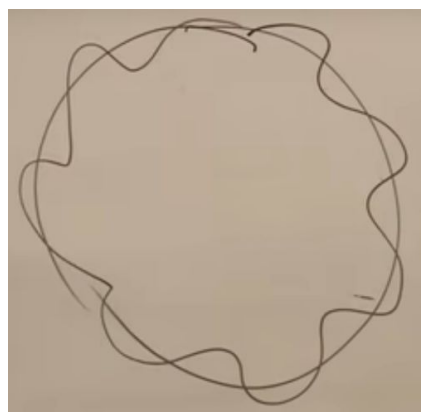
To compute radiation density, take a box, volume a^3, and assume a certain number of photons.

Figure 9: Unit box with photons: λ increases as box expands adiabatically

(a) As box expands adiabatically, photons do work on walls of box. Number of photons does not change.



(b) An analogy with standing waves. Number of nodes (zero crossings) is an adiabatic invariant: it has to be an integer. So Λ stretches with circle.

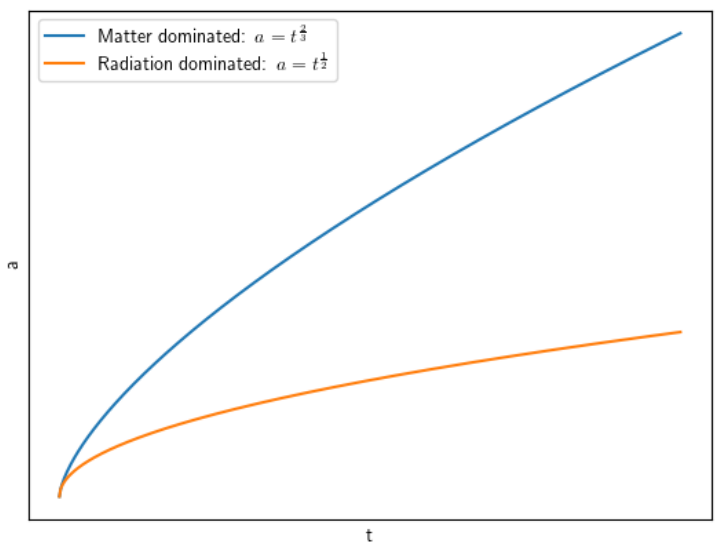


$$V = a^3$$
$$E = \frac{hc}{\Lambda}$$

We have to do some more quantum mechanics of classical electromagnetism to justify the statement in Figure 9. For the time being, take it as a given. So energy in box decreases: $E \propto \frac{1}{a}$ and $\rho \propto \frac{1}{a^4}$. We substitute in (13):

$$
\begin{aligned}
(\frac{\dot{a}}{a}) =& \frac{8\pi\nu G}{a^4} - \frac{C}{a^2}, \text{ and explore the case where } C = 0 \\
(\frac{\dot{a}}{a}) =& \frac{8\pi\nu G}{a^4}, \text{ or, with suitable units} \\
(\frac{\dot{a}}{a}) =& \frac{1}{a^4}, \text{ taking square root and rearranging} \\
\dot{a} =& \frac{1}{a} \\
\frac{dt}{da} =& a \\
t =& a^2, \text{ in suitable units} \\
a =& \sqrt{t} \qquad (14)
\end{aligned}
$$

Figure 10: Comparison of radiation and matter dominated universes–(11) and (14)



2.3 Mixed case: neither matter nor radiation dominates

We consider a real universe with both matter and radiation.

$$
(\frac{\dot{a}}{a})^2 = \underbrace{\frac{C_m}{a^3}}_{\text{More important for big } a} + \underbrace{\frac{C_R}{a^4}}_{\text{More important for small } a}
$$

At start of expansion, radiation dominates, $t^{\frac{1}{2}}$, but later switches, $t^{\frac{2}{3}}$–Figure 10.

- How do we know that there isn't conversion of matter to energy as universe expands? In practice mostly conserved separately once things cool to 1000K.
- There is a third component: dark energy (dark matter is included in C_m).
- Radiation includes photons, gravitons, and neutrinos, which move at speeds close to c.

This is the cosmology of 30 years prior to date of lecture (2013).
What happens to radiation energy lost during expansion?

$$\underbrace{-\left(\frac{\dot{a}}{a}\right)^2}_{\text{Kinetic Energy}} + \frac{C_m}{a^3} + \frac{C_R}{a^4} = 0$$

Kinetic Energy is tiny at present.
There are two ways to think about time translation symmetry.

1. The universe is a time dependent thing that everything moves in. There is no longer time translation symmetry, so energy doesn't have to be conserved.
2. There is time translation symmetry: starting universe at $t = -7$ is exactly that same as starting at $t = 4$, but this adds an extra term to energy equation. This is very subtle and we'll discuss later.

There are three cases in (13).

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi\rho G}{3} - \frac{\kappa}{a^2}$$

In Section 3, we will see that κ is connected with geometry.

- If $\kappa = 0$, space is flat.
- If $\kappa > 0$, space is curved like a sphere, and it compact.
- If $\kappa < 0$, space is curved negatively.

3 Geometries of space: flat, spherical, hyperbolic

As a matter of observation, space *is* flat. We don't know that it is flat, just that it is very big, and we don't know whether there is any curvature. It is important to investigate the various possibilities. We don't know what is happening on

scales beyond 10 billion light years, but we will make some assumptions. We won't assume that it is flat, but we will assume that it is homogeneous and isotropic. Is that true? We don't know, but we can only test it if we know its consequences. Almost all cosmology is based on that assumption.

If space is not flat, it must be curved. While there are many types of curved spaces, most are not compatible with the universe being homogenous.

- A parabaloid is most certainly not homogeneous. Its curvature is greater at the tip, less further away.
- A long pointy ellipsoid is not homogeneous. You would notice the difference if you were walking around on it.
- Surfaces with bumps are not homogeneous.

What kinds of spaces *can* be curved and homogeneous? There are two kinds, positive curvature, Section 3.1, and negative, Section 3.2.

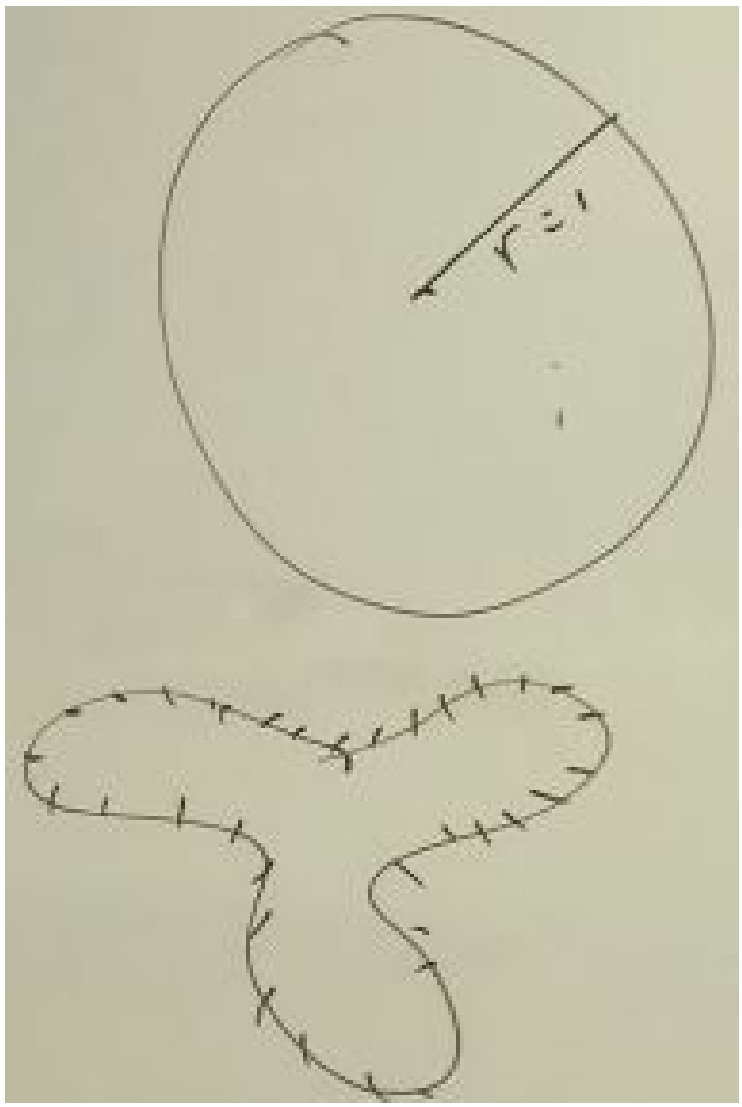
3.1 Flat Space and the 3-sphere

Let's think about a metric for ordinary flat space, restricting to 2 dimensions.

$$\begin{aligned} ds^2 =& dx^2 + dy^2, \text{ or changing to polar coordinates} \\ =& dr^2 + r^2 d\theta^2, \text{ or changing notation} \\ =& dr^2 + r^2 d\Omega_1^2 \end{aligned} \tag{15}$$

Use the idea that $d\theta^2$ is a metric for Ω_1, the 1-sphere or unit circle.

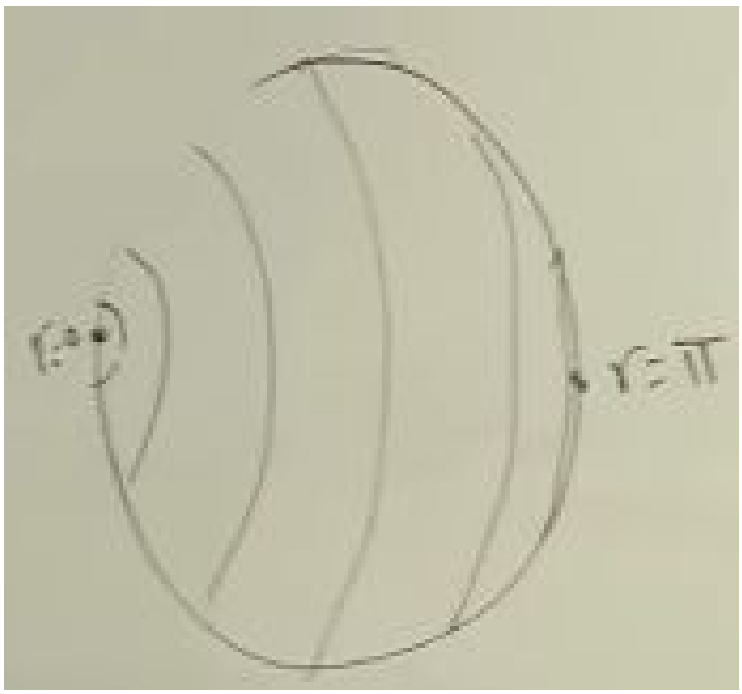
Figure 11: We can treat circle as a piece of string and deform it. Label by angular coordinate so equal distances map to the equal angles. A circle is just a curve where we come back to the same place after an angle of 2π.



In polar coordinates, it *looks* as if there is a special place, but, in reality, all points are the same. We will think of flat 2d space as a nested collection of circles–(15).

The 2-sphere is also homogeneous. Every point on the surface of the sphere is the same as any other point. It has uniform curvature. If you walk around you come back to the same space.

Figure 12: Nested spheres with point indicating observer. Observer looks out and sees a series of nested spheres, maybe one light year apart. They start growing, slow down, and then contract. Characterize spheres by angle r. We will denote the 2-sphere Ω_2.

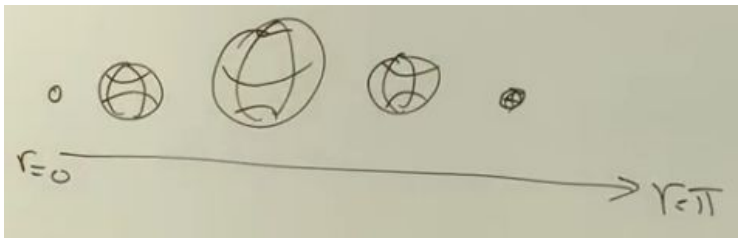


The metric of the 2-sphere is:

$$d\Omega_2^2 = dr^2 + \sin^2 r \; d\Omega_1^2$$

This pattern just continues. If I want to make a 3 dimensional sphere–a 3 dimensional space where every point is the same, and where we eventually come back to the same point. If we look into space for a certain distance, things form a 2-sphere. Look further, a bigger 2-sphere, but eventually they start to get smaller.

Figure 13: Nested 2-spheres in a 3-sphere. The observer is in the 3-sphere, at the point $r = 0$. The 2-spheres surround the observer.



The metric follows the familiar pattern:

$$d\Omega_3^2 = dr^2 + \sin^2 r \; d\Omega_2^2$$

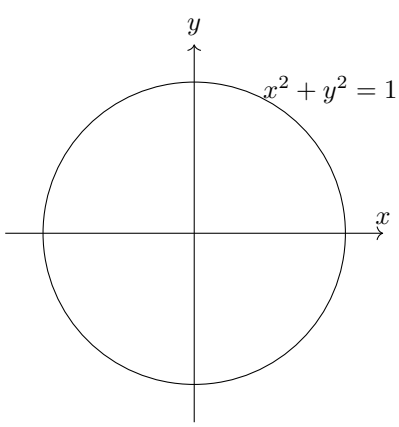
What happens in flat space? In polar coordinates, (15) generalizes to:

$$ds^2 = dr^2 + r^2 d\Omega_2^2$$

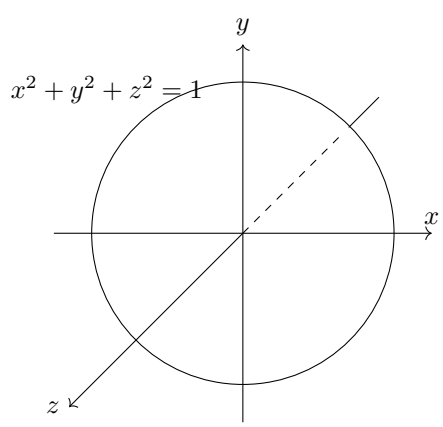
There is another way to view spheres. Imagine an ant living on a circle. It can receive light along the circle, it can communicate with its neighbours, but it has no way of knowing whether it lives on either curve in Figure 11, or even whether there are additional dimensions. We, however, can describe a circle by embedding it in two dimensions.

Figure 14: Embedding an n-sphere in $n + 1$ dimensions

(a) A 1-sphere in two dimensions



(b) A 2-sphere in three dimensions



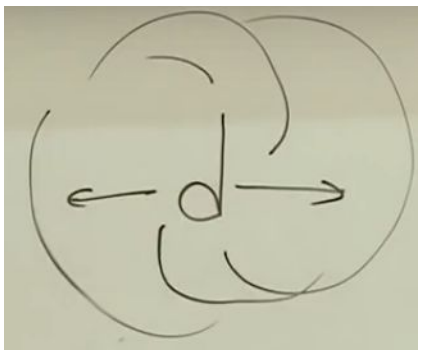
Similarly we can embed a 3-sphere in 4 dimensions: $x^2 + y^2 + z^2 + w^2 = 1$. We don't have to answer the question whether this makes physical sense.

What is the difference in what we perceive between a sphere and an infinite flat plane? Suppose we had telescopes that allow us to determine the distance

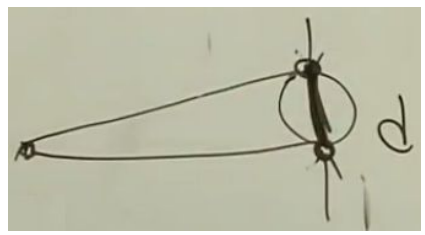
to distant objects. The telescope doesn't measure r, but there are tricks, e.g. spectroscopy, brightness(assume galaxies the same). We ask what angle they subtend.

Figure 15: Standard Galaxies in flat space

(a) Standard galaxy has width d

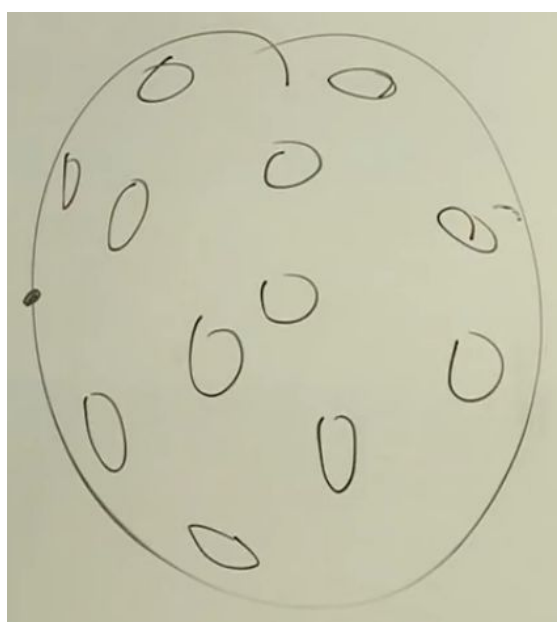


(b) For flat space, the observer perceives the width of the galaxy to be $ds^2 = r^2$, so $d\theta^2 = d^2$



Now, consider the 2-sphere.

Figure 16: Galaxies embedded in 2-sphere, with observer at "left pole".

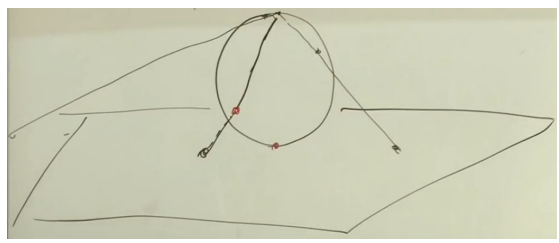


$$
\begin{aligned}
d^2 =& \sin^2 r d\theta^2 \\
d\theta =& \frac{d}{\sin r}
\end{aligned}
$$

$d\theta$ is bigger than if we lived in flat space. Distant galaxies appear bigger, but fainter. So we can distinguish flat and spherical.

We can also count the number of galaxies: on a sphere you see fewer galaxies at great distances, compared to the plane. At antipodes we see just one galaxy.

Figure 17: Stereographic Projection: observer is at South pole. Every point on sphere projects to a point on plane, and North pole projects to infinity. Circles on sphere map to little closed curves on plane (actually circles).



How do circles on the plane map to circles on the sphere? If they are close to South pole they look the same. As we move further the projections get bigger, and the extreme Arctic circle looks very large. Sphere is homogeneous, but this doesn't match plane.

We can do the same thing with a 3-sphere.

3.2 Hyperbolic geometry

There is a third homogeneous geometry, which we'll call the hyperbolic space. As we move from observer, size of circles blows up very rapidly.

$$
\begin{aligned}
d\mathcal{H}_2^2 =& dr^2 + \sinh^2 r d\Omega_1^2 \\
d\mathcal{H}_3^2 =& dr^2 + \sinh^2 r d\Omega_2^2, \text{ candidate geometry for our space.}
\end{aligned}
$$

What happens to angle subtended by a galaxy?

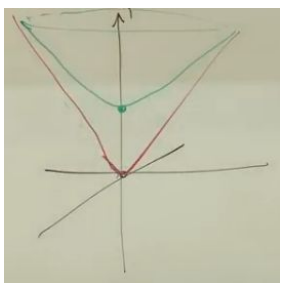
$$
\begin{aligned}
d\theta =& \frac{d}{\sinh r} \\
\approx& \frac{2d}{e^r}
\end{aligned}
$$

So if we lived in a hyperbolic universe, we'd notice that distant galaxies looked to small, and there were a lot of them.

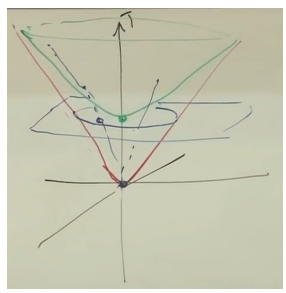
We will construct the hyperbolic surface $T^2 - x^2 - y^2 = 1$.

Figure 18: Hyperboloid.

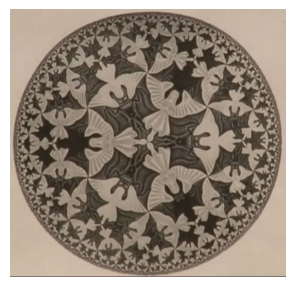
(a) This doesn't appear *homogeneous*, but is if we use metric $t^2 - x^2 - y^2$ (Lorentz Transformation)



(b) Stereographic projection. We project points onto plane. Notice asymptotic circle: there are no points outside this. There is a lot of distortion.



(c) The angels and devils are supposed to be the same size, and each one sees the same things, and has the same neighbours. There are coordinate transformation that will map one onto the other.



This geometry has many names. It is also the uniformly negatively curved space.

We can also make the space have a radius other than 1.

$$\begin{aligned} ds^2 =& a^2\big(dr^2 + \sin^2 r d\Omega_{n-1})^2\big)\text{sphere} \\ ds^2 =& a^2\big(dr^2 + \sinh^2 r d\Omega_{n-1})^2\big)\text{hyperboloid} \end{aligned}$$

The smaller a, the greater the curvature.

Modern cosmology assumes that the space we live in is one of the three that we have considered. On scales that we can currently detect, the Universe appears flat–out to 22 billion light years, so we know that the radius must be at least 10 times this side.

Space could also be a torus, a rectangle with opposite sides identified; flat, with periodic boundary conditions. People have looked for evidence of periodicity, but not found any[2].

3.3 Space and Time

The Minkowski metric is:

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$$

[2]This was in response to a question asked at the start of Section 4

A light ray (null ray) is ds^2=0–$dx = \pm dt$

We will replace $dx^2 + dy^2 + dz^2$ with one of the metrics we have discussed, e.g.

$$\begin{aligned} ds^2 =& - dt^2 + a(t)^2 d\Omega_2^2 \\ D =& a(t)\theta \\ V =& \dot{a}\theta \\ =& \frac{\dot{a}}{a} D, \text{ Hubble Law} \end{aligned}$$

Space is 2 dimensional, and diameter of space changes with time, $2 + 1$

Here are our 3 cases of interest:

$ds^2 = - dt^2 + a(t)^2 \left(dx^2 + dy^2 + dz^2\right)$, Flat, but scale factor depends on time

$ds^2 = - dt^2 + a(t)^2 d\Omega_3^2$, spherical, with varying radius

$ds^2 = - dt^2 + a(t)^2 d\mathcal{H}_3^2$, hyperbolic, with varying radius

We need to investigate how a varies with time. We will need to write down equations from General Relativity. It turn out they will agree with the Friedmann equations we derived from Newtonian physics.

4 Cosmological Thermodynamics

4.1 Questions from audience

4.1.1 if universe is expanding, how come planets maintain stable orbits?

Hold hands out in space with friend. There is a very mild repulsion from expansion, but repulsion is dwarfed by holding hands. Also atoms closer together, so attraction dominates Hubble attraction. Consider at atom, which is also embedded in space. Electrostatic forces overwhelm expansion. Atom will be expanded a tiny bit, but not enough to fly apart. Same is true of the solar system: gravitational pull of Sun overwhelms tendency to expand.

Energy conservation in an expanding universe is different from energy conservation in a static universe. The simplest way to look at is is to note that energy conservation is a consequence of time translation invariance: if everything is time invariant, we get energy conservation. If background is not static, energy conservation does not apply. Nett amount of energy isn't fixed: changes in energy translate into kinetic energy of expansion.

NB: Homogeneity and isotropy is on a large scale; things are not homogeneous out to a few hundred million light years.

4.1.2 Clarification of homogeneity

Homogeneity means that it is possible to write the metric so that it has the same form anywhere. Consider 2 dimensional flat space:

$$\begin{aligned} ds^2 =& dx_1^2 + dx_2^2. \text{ Suppose we translate the coordinates} \\ \vec{y} =& \vec{x} + \vec{a}. \text{ Then in the new coordinates} \\ ds^2 =& dy_1^2 + dy_2^2, \text{ which has the same form.} \end{aligned}$$

We can show that a similar thing happens with $ds^2 = dr^2 + \sin^2 r d\theta^2$ if we move observer from South pole to East pole. In 2 dimensions it is enough that the curvature be the same at every point. The hyperbolic surface is also homogeneous and isotropic. The torus is also translation invariant, but not isotropi.c, because of there being preferred directions.

Being homogeneous and isotropic is somewhere between being a postulate and an observational fact.

Geometry has its own intrinsic shape, it isn't necessarily embedded. In 1 dimension the only thing that matters is distance along the curve. And they are not curved: we straighten any piece. They are flat but closed.

What is special about 3 dimensions? Our brain architecture evolved to navigate a 3 dimensional space. Mathematically 3 dimensional space isn't special.

4.2 Setting up the Geometry

Let us assume that space has one of the 3 homogeneous and isotropic geometries described in Section 3.3.

General Relativity starts with a metric, and we will assume that space and time aren't mixed with each other in the metric[3]. The metric has one of these three forms:

$$\begin{aligned} ds^2 =& -dt^2 + a(t)^2 d\Omega_3^2 && \kappa = 1, \text{ or} \\ ds^2 =& -dt^2 + a(t)^2 \left(dx^2 + dy^2 + dz^2\right) && \kappa = 0, \text{ or} \\ ds^2 =& -dt^2 + a(t)^2 d\mathcal{H}_3^2 && \kappa = -1 \end{aligned}$$

In all three cases:

$$\begin{aligned} D =& a(t) d\theta \\ V =& \dot{a} d\theta \\ =& \frac{\dot{a}}{a} D \end{aligned}$$

Hubble's parameter $\frac{\dot{a}}{a}$ depends on time, but not on position.

[3]This is a consequence of homogeneity and isotropy

4.3 Einstein's Field Equation

From [1, General Relativity, Lecture 9]:

$$\underbrace{R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R}_{\text{Einstein Tensor}} = \underbrace{\frac{8\pi G}{3}}_{\text{c.f. equation (12)}} \underbrace{T^{\mu\nu}}_{\text{Energy-momentum}} \tag{16}$$

$T^{\mu\nu}$ contains a complex of things: the density of energy. the flux of energy, the density of momentum, and the flux of momentum. We will fix on the time-time component[4], $\frac{8\pi G}{3}T^{00}$. T^{00} is the energy density, ρ, so we will write $\frac{8\pi G}{3}\rho$. The right hand side is completely sensitive to the type of material that is in the universe, e.g. matter, photons: it knows about the material that makes up the universe.

The curvature has two parts, one of which involves second derivatives of the metric by the coordinates, the other squares of first derivatives. Fortunately the time-time component contains only one of the two: the second derivatives cancel, and we only have squares of first derivatives.

There are derivatives by time, and by space. The space ones are connected with curvature of space, which scales as $\frac{1}{R}$; the time ones with how things are changing with time, i.e. $a(t)$. Einstein's Field Equation, (16), reduces to:

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 + \frac{\kappa}{a^2} =& \frac{8\pi}{3}\rho, \text{ or} \\ \left(\frac{\dot{a}}{a}\right)^2 =& \frac{8\pi G}{3}\rho - \frac{\kappa}{a^2}, \text{ c.f. Equation (13)}\end{aligned} \tag{17}$$

We can interpret κ as curvature, rather than energy.

Why does General Relativity give the same answer as Newton? If we just look at a small region of curved space, it looks flat. Once we realize that equations are the same for a small piece and a large one, we see that we must reproduce Newton's equations. In Newton's view, ρ is the mass density, which if accurate if everything is moving at speeds much less than light. On the other hand, the Einstein equations are more general than Newton's: they apply to fast moving particles, and photons.

4.4 Equation of State

We need to understand how ρ depends on a to solve (17). We will consider two cases:

$$\rho = \frac{\rho_0}{a^3}, \text{ Matter dominated, and} \tag{18}$$

$$\rho = \frac{\rho_0}{a^4}, \text{ Radiation dominated} \tag{19}$$

There are also combinations of these two.

We will proceed in two steps:

[4] This turns out to give Newton's energy equation; other components would give $F = mA$.

1. Assume an equation of state, and use it to derive equations like (18) and (19).

2. Derive equation of state for different kinds of materials.

An equation of state is a relationship between thermodynamic variables describing a system. For our system, temperature won't play a big role. We will be interested in the relationship between energy density and pressure. The example of interest can be described by very simple equations of state.

$$\begin{aligned} P =& w\rho, \text{ for some constant } w \\ w =& 0, \text{ for a matter dominated universe} \\ w =& \frac{1}{3}, \text{ radiation dominated–see (25)} \end{aligned} \tag{20}$$

Consider a box of gas. We can work out the total energy from the density and volume, V.

$$E = \rho V \tag{21}$$

We change the volume, doing work. From [1, Statistical Mechanics, Lecture 6], assuming an adiabatic change:

$$\begin{aligned} dE =& -PdV + \underbrace{TdS}_{=0} \\ =& \rho dV + Vd\rho, \text{ from (21). Combine with (22)} \\ Vd\rho =& -\big(p+\rho\big)dV \end{aligned} \tag{22, 23}$$

(It is permissible to add p and ρ as the latter is *energy* density. Both have dimension $\mathsf{T}^{-2}\mathsf{L}^{-1}\mathsf{M}$.)

$$\begin{aligned} Vd\rho =& -(1+w)\rho dV, \text{ from (20) and (23). Regrouping} \\ \frac{d\rho}{\rho} =& -(1+w)\frac{dV}{V}, \text{ and integrating} \\ \rho =& \frac{C}{V^{1+w}} \\ =& \frac{C}{a^{3(1+w)}} \end{aligned} \tag{24}$$

- If $w = 0$, we have a matter dominated universe, with $\rho = \frac{C}{a^3}$
- If $w = \frac{1}{3}$, we have a radiation dominated universe, with $\rho = \frac{C}{a^4}$

We need little more than the equation of state to derive a universe.

4.5 Dark Energy, Cosmological Constant, Vacuum Energy

Let's try $w = -1$. (E.g Replace box with a line interval with a spring joining the ends. Pressure is negative[5], as spring is drawing ends together. Increasing potential energy makes pressure more negative.) Then ρ is constant: it does not change when box expands.

We can mock up the cosmological constant by adding linear term to Newton's gravitational law.

5 Vacuum energy

What is the history of the Universe? To the extent that it homogeneous and isotropic, this question boils down to the time history of the scale factor. If we know the history of the scale factor we know an awful lot about the history of the universe. We can test it, and observe it.

We have looked at

- the matter dominated universe, in which $a(t) \propto t^{\frac{2}{3}}$–Equations (11)&(18),
- energy dominated $a(t) \propto t^{\frac{1}{2}}$–Equations (14) & (19).

Both are models, and neither is absolutely correct. We believe that the second model was approximately correct for the early universe, and that the first became true later. We will discuss when we get to observational cosmology.

We also looked at the relation between a and ρ–Equation (24).

- $\rho = \frac{\rho_0}{a^3}$ (matter dominated),
- $\rho = \frac{\rho_0}{a^4}$, radiation dominated.

The equation of state is $P = w\rho$–Equation (20). In a matter dominated universe, because $E = mc^2$, a tiny amount of mass still has a large energy density. But particles have a low velocity; where does pressure come from? From particles hitting the wall, slowly; so formula for pressure contains velocities of particles, but not c. It is much less than energy density, so $w = 0$ for a non-relativistic matter dominated universe. For radiation, $w = \frac{1}{3}$, which we will prove in Section 5.1.

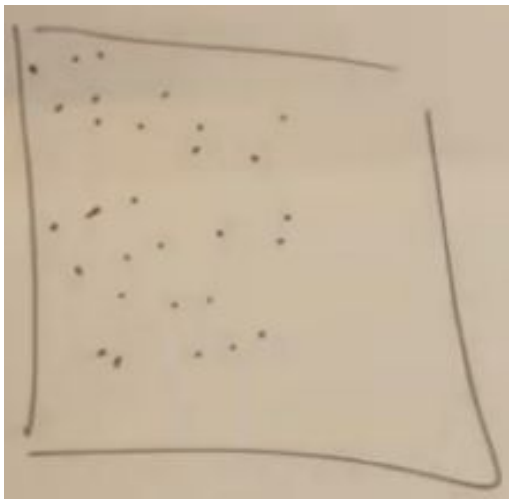
5.1 Why is $w = \frac{1}{3}$?

Radiation is massless particles–photons. Figure 19a depicts a box of photons.

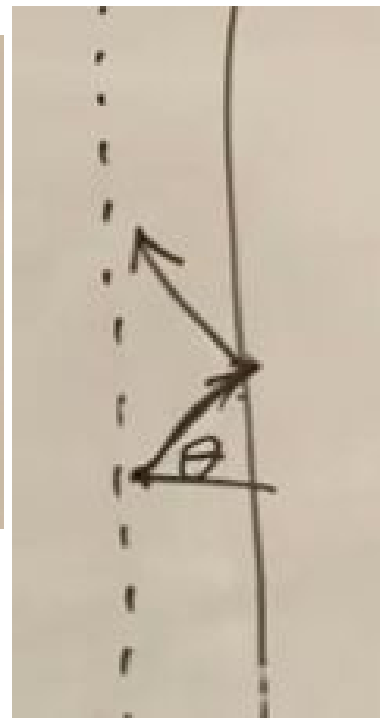
[5] We call this tension

Figure 19: A box of photons

(a) Photons bounce of walls, lose no energy, and exert pressure on walls



(b) Photon hitting wall at angle θ



Let us pretend that all photons have the same energy (we will find out this doesn't matter; it is approximately true of photons are in thermal equilibrium). Let:

- energy per photon be ϵ;
- momentum per photon $\vec{\pi}$, (not "the" π), and
- number of photons per unit volume be ν.

In any little volume of box, on average momentum could be in any direction (isotropic).

$$\begin{aligned}\epsilon =& \pi c, \text{ where}\\ \pi =& |\vec{\pi}|, \text{ so}\\ \epsilon =& \pi, \text{ in units where } c = 1\end{aligned}$$

Let number of photons per unit volume be ν. Then the energy density is:

$$\rho = \nu\epsilon$$

How many particles hit the box during time interval Δt? In Figure 19b, a particle moving to right will hit the wall during Δt if it is close enough–within ΔX–where $\Delta X = \Delta T$. What if it is not moving exactly right, say at angle θ–Figure 19b?

$$\begin{aligned}
\Delta x =& \Delta t \cos\theta. \text{ Then the momentum will change by:} \\
\Delta_\pi =& 2\epsilon\cos\theta, \text{ since particle turned around, so} \\
\frac{\Delta_\pi}{\Delta t} =& \frac{2\epsilon\cos\theta}{\Delta t}, \text{ is the force per particle that hits wall}
\end{aligned}$$

Now the number of particles that hit the wall is given by

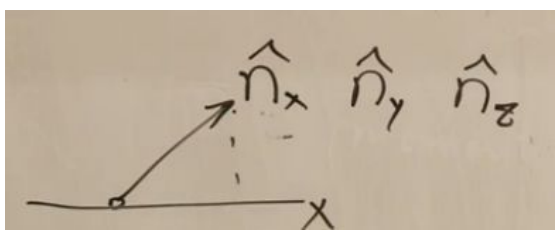
$$N = \frac{1}{2}\Delta x A \nu, \text{ where } A \text{ is area, and } c = 1$$

allowing for the fact that half the particles within Δx are moving away. We now calculate the contribution to the pressure from particles moving at angle θ .

$$\begin{aligned}
P(\theta) =& \frac{1}{2}\frac{2\epsilon\cos\theta}{\Delta t}\Delta x\ \nu, \text{ force per area} \\
=& \frac{\cancel{2}}{\cancel{2}}\epsilon\cos\theta\Delta x\nu, \text{ force per area} \\
=& \epsilon\nu\cos^2\theta \\
=& \rho\cos^2\theta
\end{aligned}$$

Now integrate over all θ. We shall suppose that every direction is characterized by a unit vector $\vec{n}$, Figure 20.

Figure 20: Unit vector $\hat{n}$



We see that:

$$\begin{aligned}
n_x =& \cos\theta \\
\hat{n}_x^2 + \hat{n}_y^2 + \hat{n}_z^2 =& 1
\end{aligned}$$

Now n_x, n_y, and n_z are related by rotational symmetry. We can take average over 3D space, and use isotropy to get:

$$\begin{aligned}\langle \hat{n}_x \rangle^2 + \langle \hat{n}_y \rangle^2 + \langle \hat{n}_z \rangle^2 =& 1 \\ P =& \frac{1}{3}\rho, \text{ or} \\ w =& \frac{1}{3}, \text{ as asserted in Section 4.4}\end{aligned} \tag{25}$$

We assumed all particles have the same energy. If not, we could interpret (25) as the contribution from a particular energy, and integrate over all energies.

What wall are we talking about? There is no wall in space. Really they are going right through, but, on average there is one coming from the opposite direction, so, on average, our argument works for thermal equilibrium (CMB is in thermal equilibrium).

5.2 Vacuum Energy

Can energy density ever be negative? Yes, but not under any circumstances we will be interested in. (Yes it can). Pressure can be negative–tension: two ends connected by a spring. If pressure negative and ρ positive, we have negative w. (LS would not bother unless it were central to cosmology.) If particles are attractive they can give negative pressure.

Vacuum energy comes from quantum field theory, and gives rise to negative pressure. It is an energy that we assign to empty space (zero point energy of fluctuation). A box contains a vacuum energy equal to vacuum energy density times volume.

The vacuum energy density is a constant; it is independent of $a(t)$:

$$\rho_0 = \Lambda \frac{3}{8\pi G}$$

Λ is known as the "cosmological constant", or as "dark energy". For the significance of the factor $\frac{3}{8\pi G}$, see (12)

We shall work out the equation of state.

$$\begin{aligned}E =& \rho_0 V \\ dE =& \rho_0 dV \\ =& -w\rho_0 dV \text{ from (20), whence} \\ w =& -1\end{aligned}$$

When you read that astronomers are measuring w, and discovering that w is close to -1, they are talking about vacuum energy. The closer the experimental evidence is to $w = -1$, they are saying that the energy of the universe is like vacuum energy; it doesn't dilute, because it is a property of empty space.

Vacuum energy can be positive or negative, but, in both cases, pressure and density have opposite sign. We don't know the value of ρ_0: there are too many contributions.

Let us study the case where the vacuum energy density is the *only* density. Earlier we look at a pure matter dominated universe, Section 2.1, and now we'll look at a pure vacuum energy universe. There are six cases for non-zero Λ and for κ–Table 1.

The Friedmann equation, (12), becomes:

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 =& \frac{8\pi G}{3}\rho_0 - \frac{\kappa}{a^2} \\ =& \Lambda - \frac{\kappa}{a^2}\end{aligned} \tag{26}$$

Table 1: Options for Λ and κ

Λ	κ	Remarks
0	+1	Already done
0	0	Already done
0	-1	Already done
+1	0	de Sitter space–see Section 5.2.1
+1	+1	de Sitter space–see Section 5.2.2
-1	+1	Makes no sense as it would make $\left(\frac{\dot{a}}{a}\right)^2 < 0$
-1	-1	Section 5.2.3

5.2.1 Flat space, positive cosmological constant

Substituting $\Lambda > 0$ and $\kappa = 0$ in Friedmann's equation (26) gives:

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 =& \Lambda \\ \dot{a} =& \sqrt{\Lambda} a \\ a(t) \propto& e^{\sqrt{\Lambda} t}, \text{ universe expands exponentially}\end{aligned}$$

The Hubble constant is $\sqrt{\Lambda}$, and $a(t) \propto e^{Ht}$. This is known as "de Sitter space".

Let us mix in some matter:

$$\left(\frac{\dot{a}}{a}\right)^2 = \Lambda + \frac{C}{a^3}$$

In early times, a is small, and $\frac{C}{a^3}$ will dominate; later, a is large and Λ will dominate. Right now the two energies compete (for last 2-3 billions years, dark energy has started to become more important. Observation shows that it is accelerating, and, as precision increases, it looks more like exponential acceleration.)

Positive vacuum energy comes from bosons, negative from fermions. LS says it is a mystery that vacuum energy is so low.

5.2.2 Positive curved space, positive cosmological constant

Substituting $\Lambda > 0$ & $\kappa > 0$ in the Friedmann equation, (12), gives:

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 =& \Lambda - \frac{1}{a^2}\\ \dot{a}^2 - \Lambda a^2 =& -1\end{aligned}$$

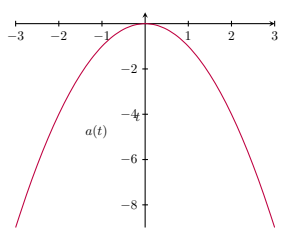
Think of $a(t)$ as the coordinate of a particle on a line. $\dot{a}^2$ becomes twice the kinetic energy; $-\Lambda a^2$ the potential energy plotted in Figure 21a. Think of a starting large, going up hill, then rolling back.

We can solve to get:

$$a(t) = \frac{1}{\sqrt{\Lambda}} \cosh\left(\sqrt{\Lambda} t\right)$$

Figure 21: $\dot{a}^2 - \Lambda a^2 = -1$

(a) Potential Energy



(b) '$a(t) = \frac{1}{\sqrt{\Lambda}} \cosh\left(\sqrt{\Lambda} t\right)$'

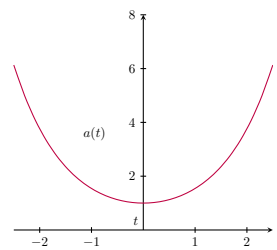
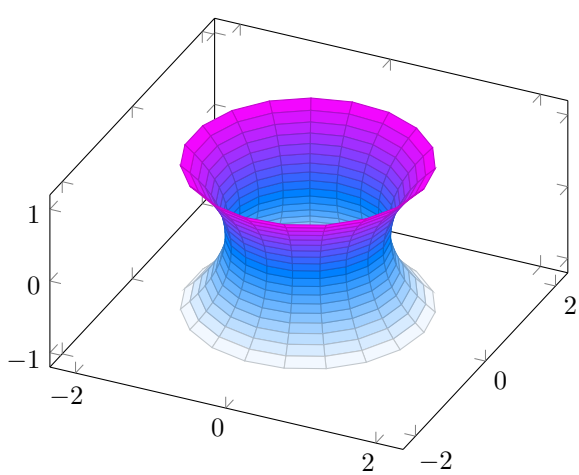


Figure 22: Universe bounces at $t = 0$



5.2.3 Negative cosmological constant, with negative curvature

We will consider a negative cosmological constant, with negative curvature:

The Friedmann equation, (12), becomes:

$$\begin{aligned}(\frac{\dot{a}}{a})^2 =& -1+\frac{1}{a^2}\\ \dot{a}^2+a^2 =& 1, \text{ the energy equation for a harmonic oscillator}\end{aligned}$$

Universe expands and crunches.

6 Dark matter and allocation of energy density

We will look at the connection between experimental facts and our models.

6.1 Review

$$\begin{aligned}(\frac{\dot{a}}{a})^2 =& H^2, \text{ Friedmann, Hubble}\\ =& \frac{8\pi G}{3}\rho-\frac{\kappa}{a^2}\end{aligned}$$

This changes with time: when people speak of the "Hubble constant" they mean today's value.

Our metric is:

$$\begin{aligned}ds^2 =& -dt^2+a(t)^2\left[dr^2+\xi^2(r)d\Omega^2\right], \text{ where} && (27)\\ \xi(r) =& \sin r && \text{if } \kappa=+1\\ \xi(r) =& \sinh r && \text{if } \kappa=-1\\ \xi(r) =& r && \text{if } \kappa=0\end{aligned}$$

ξ depends on the geometry.

There are only 3 contributions to ρ known.

$$-\frac{\kappa}{a^2}+\frac{8\pi G}{3}\rho = \underbrace{\frac{C_R}{a^4}}_{\text{Radiation}} + \underbrace{\frac{C_M}{a^3}}_{\text{Matter}} + \underbrace{\Lambda}_{\text{Vacuum energy}} - \frac{\kappa}{a^2} = H^2 \qquad (28)$$

Equation (28) is a constraint. Each term has some value, as does H. How do we measure them?

- Hubble determined his constant by plotting red-shift against luminosity. He miss-measured by a factor of 10. We do better today, and we look at a small enough part of universe that we don't have to worry about age varying.

- What about C_R? We can look at the amount of radiation in universe, mostly in CMB, but it isn't very big.
- Matter, we look at the galaxies, and see how much hydrogen is in them, measure luminosity, and measure luminous matter. It comes to one proton per cubic metre. Low, but much higher than radiation density.
- Forget vacuum energy for the time being. We are being historical, so leave out dark matter and dark energy.

Today we call the ratios Ω_R, Ω_m, Ω_Λ, and Ω_κ,normalized so:

$$\Omega_R + \Omega_m + \Omega_\Lambda + \Omega_\kappa = 1$$

When LS was a student, Ω_R was negligible, Ω_Λ was unknown, and Hubble constant was 20 times larger than Ω_m, so it was assumed that $\Omega_\kappa \approx 1$. The meant on open, infinite universe with $k \approx -1$. For most of the time $\frac{c_M}{a^3}$ competes favorably with $\frac{\kappa}{a^2}$

H has another meaning in a matter dominated universe, where $a \propto t^{\frac{2}{3}}$. Also:

$$\begin{aligned} \frac{\dot{a}}{a} =& \frac{2}{3t} \\ H =& \frac{2}{3t} \\ H_{today} =& \frac{2}{3T} \\ \approx& 10^{10} \text{ years} \end{aligned}$$

We also have (From where?)

$$\begin{aligned} \frac{1}{a^2} =& \frac{1}{H^2}, \text{ so} \\ a \propto& T \end{aligned}$$

None of this is true: it is what they thought 40 years ago.

6.2 Dark Matter

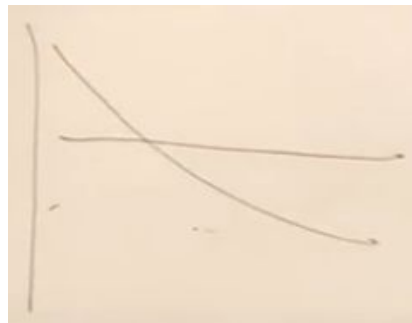
Ordinary matter is made from luminous matter: it radiates when it gets hot. In 1932 Oort noticed some galaxies were misbehaving, and, the following year, Zwicky noticed some clusters of galaxies were misbehaving: things were not moving the way Newton's laws predicted, unless there was a great deal of mass that was not accounted for. Black holes aren't massive enough; moreover, Oort realized that the extra mass could not be concentrated at the centre.

Most of the luminous matter is in the centre of the galaxy, so the outer parts should move under a central force.

Figure 23: Most of the luminous matter is in the centre of the galaxy, so the outer parts should move under a central force.

(a) A star moving under a central force

(b) Observed velocity is constant, unlike predicted velocity



$$\frac{MG}{r^2} = \frac{v^2}{r}$$

$$v^2 \propto \frac{1}{r}, \text{ Kepler}$$

But this isn't what is observed–Figure 23b: velocity (not angular velocity) is pretty much constant. Newton would have assumed mass is distributed in 3D in accordance with some function $M(r)$

$$\frac{M(r)G}{r^2} = \frac{v^2}{r}$$

$$\frac{M(r)G}{r} = v^2, \text{ but } v \text{ is constant, so}$$

$$M(r) \propto r$$

6.2.1 Dark Matter

So M, not ρ, increases linearly, at least out to the boundary of the galaxy. LS does not know of a good argument why mass behaves this way. Observation of stars to top and bottom of galaxy suggests that distribution is spherical, and it is ten times mass of luminous mass.

Why doesn't it collapse? Luminous matter lost energy by radiation. Dark matter isn't changed, and particles are very light.

LS suggest that dark matter clumped first, then baryons were clumped also and lost energy through radiation. Black holes didn't play [much of a] part.

Why not neutrinos? They are so light they'd need to move very fast ("hot dog matter"), but dark matter tends to cluster.

Dark matter didn't make the discrepancy with the Hubble constant go away, but it lessened it.

Dark matter can't be fermions if they are light, but maybe bosons. Bose condensate? (garbled!)

6.2.2 Modify gravitational theory

The alternative is to modify gravitational theory Newton's laws to that Kepler's 2nd law doesn't work for galaxies, but this doesn't seem to work. Most physicists think it is particles. Since Newtonian gravity works find in our solar system, we know that ordinary matter dominates here. Matter in solar system is much denser than interstellar space.

6.3 Testing cosmological theories

How do we test the following equation?

$$\frac{C_M}{a^3} - \frac{\kappa}{a^3} = H^2$$

A model consists of κ and an equadrtion of state, for which we can substitute a history, $a(t)$. We can measure red-shift, and luminosity.

To start, assume we can ignore time. how can we measure κ? Meausure number of galaxies of a given brightness, as a function of distance.

- In a flat space, the number of galaxies in a shell of thickness dr goes as r^2.
- In a hyperbolic space, the number of galaxies in a shell of thickness dr goes as $\sinh^2(r)$, which is exponential, $e^{2r}dr$
- In a 3-sphere, the number of galaxies in a shell of thickness dr goes as $\sin^2(r)$

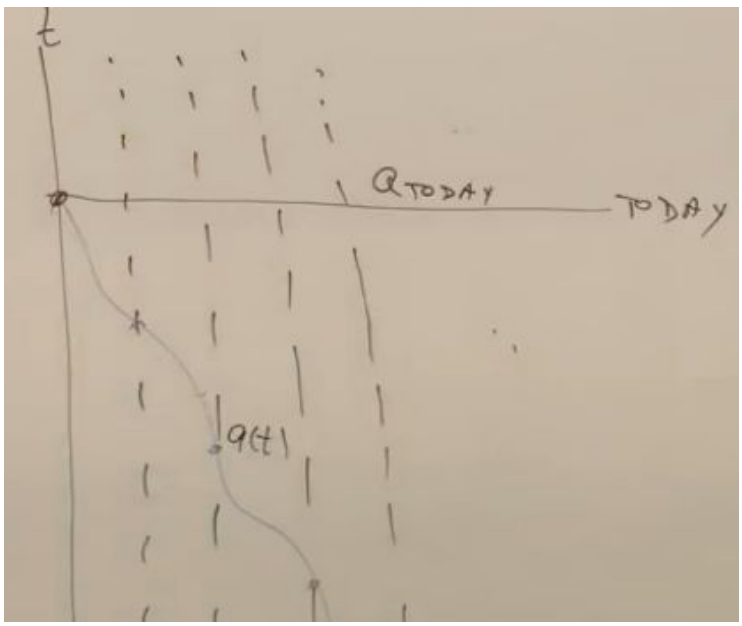
The evidence favours a hyperbolic space, as a function of red-shift. What about luminosity?

- In flat space, luminosity falls of as $\frac{1}{r^2}$
- In hyperbolic space, luminosity falls mush faster.

What about number of galaxies as a function of red-shift?

Let $\Lambda_e mitted$ denote the wavelength of light is emitted from a source. This may or may not be the wavelength that is detected: Doppler shift is one possible reason; gravitational shift due to a black hole is another; expansion of the universe is equivalent to the Doppler shift, but it is what we are interested in.

Figure 24: Map of the Universe. Vertical lines correspond to equal values of r, wavy line is light ray, and we see a number of galaxies. They have different scale factors, $a(t)$: c.f. a_{TODAY}. Light emitted at point with $a(t)$ has to stretch to get to where we observe it, by a factor of $\frac{a_{TODAY}}{a_t}$.



Definition 2 (Redshift)**.**

$$\begin{aligned} z =& \frac{\Lambda_{Detected}}{\Lambda_{Emitted}} - 1, \; -1 \; \textit{for historical reasons} \\ =& \frac{a_{TODAY}}{a_t} - 1 \end{aligned} \tag{29}$$

Each galaxy we observe at a different distance was emitted at a different time, and has its own redshift. But until we know how the light ray moves, we don't know how to compare scale factors.

A light ray is a null ray for the metric (27). If a light ray is moving radially towards us, $dt^2 = a(t)^2 dr^2$. This has two solutions, one going back into the past (the one we are interested in), and one going to the future. The past solution is:

$$
\begin{aligned}
dt =& -a(t)dr, \text{ as } t \text{ gets more negative, } r \text{ get bigger} \\
dr =& -\frac{dt}{a(t)} \qquad (30)
\end{aligned}
$$

Let us invent something an astronomer can test: what is $\frac{dN}{dz}$, the number of galaxies per unit redshift.

$$
\begin{aligned}
\frac{dN}{dz} \propto& \frac{dr\xi(r)^2}{dz} \\
dz =& -\frac{a_{TODAY}}{a(t)^2}da \\
\frac{dN}{dz} \propto& \frac{dr\xi(r)^2}{-\frac{a_{TODAY}}{a(t)^2}\dot{a}dt} \\
\propto& \frac{\xi(r)^2}{-\frac{a_{TODAY}}{a(t)^2}\dot{a}}\frac{dr}{dt} \\
\propto& \frac{\xi(r)^2}{\frac{a_{TODAY}}{a(t)}\dot{a}}, \text{ from (30)} \\
\propto& \frac{\xi(r)^2}{(z+1)\dot{a}}, \text{ from (29)} \qquad (31)
\end{aligned}
$$

There is nothing we can do with this until we make a model. We will take the matter dominated universe, and set $\kappa = -1$.

$$
\begin{aligned}
a \propto& t^{\frac{2}{3}} \\
\dot{a}\frac{2}{3}t^{\frac{1}{3}}
\end{aligned}
$$

Now a, $\dot{a}$, r can all be expressed in terms of z, so we can express the right hand side of (29) in terms of z (only because we have a model). Now $\xi(r)$ increases exponentially, which is easy to detect: it makes an anomalously large galaxies[6] at large redshift. You can convert a κ and equation of state[7] into a prediction for the number of galaxies at a given redshift. Given this, and the relationship between luminosity and redshift we can work out the complete history of the universe, and determine C_R.

[6] Galaxies is used to mean standard candles, and the best ones are supernovae

[7] The Ωs form part of the model.

Observations over the last 50 years are inconsistent with matter dominated universe with negative κ. Best fit today is:

$$\begin{aligned}
\Omega_R =& 0 \\
\Omega_m \approx& 0.3 \text{ including dark matter} \\
\Omega_\Lambda \approx& 0.7 \\
\Omega_\kappa \approx& \pm\, 0.01
\end{aligned}$$

If you vary these numbers by more than one percent, model fails to match data. Also we can determine Ω_R and Ω_m (subject to assumption about dark matter), so we only need to search on other two.

The values above are today's values:

- Long in the past $\Omega_R \approx 1$;
- long in the future $\Omega_\Lambda \approx 1$

We get a $\pm 10\%$ model with supernova data; add in the CMB we get $\pm 1\%$.

7 Temperature History of the Universe

7.1 Review

Last week we saw that the best way to estimate the rough parameters of the expanding universe is to begin with a model, then calculate its consequences; if that doesn't work so well, adjust parameters and keep going until we agree with data. The model consists of a time history for the scale factor, and a specification for the geometry. Until we get to the CMB and its details, we typically don't see deep enough and far enough that the κ is critical to our analysis. Counting galaxies and supernovae with redshifts that are no bigger than 7, they aren't sensitive to curvature. That is both good and bad: it is bad because we can't deduce the curvature; but since it isn't sensitive to curvature, we don't need to know its value to work out the history of the scale factor. We can work out the relationship between brightness and distance, and the number of objects with a given redshift. Data is consistent with negligible curvature and about 70% dark energy.

We have to look deep into Cosmic Microwave Background to get more.

7.2 Black Body Radiation

Temperature is a feature of thermal equilibrium. Thermal equilibrium is a condition for a system that is established after a period of time by scattering.

Let's imagine a box with perfectly reflecting walls and put photons into it. Open wee hole, shine laser in, then close hole. If photons don't interact, box won't come into thermal equilibrium. We need *scattering* to get thermal

equilibrium. Charged particles will scatter effectively. If box has some electrons, photons will scatter, electrons will scatter, and particles will eventually form a thermal soup; we could say that the box has a temperature.

The universe appears to be electrically neutral: $N_{e^-} = N_P$. Hydrogen atoms aren't effective scatterers, so today the universe is not in thermal equilibrium, or the different kinds of particles aren't in thermal equilibrium with each other. If electrons and protons weren't bound, they would be efficient scatterers, and the universe would have a well defined temperature. We need a high temperature to ensure they aren't bound.

Returning to our box, we have a temperature T, and each photon has an wavelength λ and frequency ν (there are many different wavelengths in the box).

Radiation is characterized by the intensity: $I(t,\nu)dVd\nu$ is energy per unit volume in band of frequencies $d\nu$. First attempt (before 1900), using dimensional analysis.

$$dim\ I(t,\nu) = [\frac{ET}{V}]$$

What are units of temperature–i.e. energy per particle? $E = k_B T$ (Boltzmann's constant is just a conversion factor: temperature can be treated as energy). We can ask what sort of dependence I can have on T and ν? The only thing that will work dimensionally is:

$$I \propto \frac{Tk_B\nu^2}{c^2} \tag{32}$$

This a a disaster: I grows without bound as ν grows, and the energy in the box is infinite. We need another constant to avoid the "ultraviolet catastrophe". The correct formula for black body radiation is:

$$I = \frac{2h\nu^3}{c^2}\frac{1}{e^{\frac{h\nu}{k_BT}} - 1} \tag{33}$$

For small ν, (33) reduces to (32), and for $\frac{h\nu}{k_BT} \gg 1$ it behaves more like $e^{\frac{-h\nu}{k_BT}}$, averting the ultraviolet catastrophe.

At the crossover point

$$\frac{h\nu}{k_BT} = 1, \text{ or}$$

$$\frac{hc}{k_BT} = \lambda, \text{ the thermal wavelength associated with temperature } T.$$

We see that the classical picture works well for longer wavelengths, but the quantum one is needed for smaller wavelengths. Most of the power is at the crossover point, at wavelengths inversely proportional to temperature. We can use this approximately even if system is not in equilibrium. At present the

wavelength as about 1mm, which corresponds to 3K: do not interpret this as saying that anything is in thermal equilibrium.

From (33):

$$I \propto \underbrace{\frac{\nu^3}{T^3}\frac{1}{e^{\frac{h\nu}{k_B T}}-1}}_{\text{Determines shape of curve}} T^3$$

If we change temperature, $\frac{\nu^3}{T^3}\frac{1}{e^{\frac{h\nu}{k_B T}}-1}$ simply rescales. This is the universal behaviour of black body radiation.

7.3 Decoupling

Most charged particles are bound into atoms, and aren't available for scattering. The universe today is not in thermal equilibrium.

When the temperature is too low in the box, the radiation decouples from the charged particles.

Imagine the box starts at a high temperature so matter is ionized. As we expand the box it cools, the temperature goes down, and the particles combine into atoms. There isn't enough oomph to ionize atoms, and the radiation decouples. The temperature of the radiation goes down, but it doesn't stay in thermal equilibrium. As the box expands, the shape of the black body radiation is frozen in. Wavelengths are stretched, which is equivalent to changing temperature, but only for radiation (Molecules have a different temperature). That is why radiation today has black body form (this has been measured). CMB has been studied, and error bars are very tiny. We see a universe filled with radiation that appears to be the remnant of a thermal distribution that must have been thermalized at a very high temperature when all matter was ionized.

$$\begin{aligned}\frac{\Lambda_{TODAY}}{\Lambda_{DECOUPLING}} =& \frac{a(TODAY)}{a(DECOUPLING)}\\ =& \frac{T_{DECOUPLING}}{T_{TODAY}}, \text{ Cosmic Microwave Background}\\ \approx& \frac{T_{ionization}}{3K}, \text{ ionization energy for hydrogen=13.5 ev}\end{aligned}$$

The ratio of photons to electrons is very large: $\frac{N_\gamma}{N_{e^-}} \approx 10^8$. (Observable fact: why?)

The probability of a photon having energy ϵ is $e^{-\frac{\epsilon}{k_B T}}$.

- What is probability that it will have the ionization energy is $e^{-\frac{\epsilon_{ion}}{k_B T}}$.
- If $\epsilon > k_B T$ this is very small, but there are many photons.

- We could ask how many photons can ionize the atom? $10^8 e^{-\frac{\epsilon_{ion}}{k_B T}}$. We are operating in tail of Boltzmann distribution.
- If $10^8 e^{-\frac{\epsilon_{ion}}{k_B T}} \approx 1$, there are enough photons to ionize the electrons. $10^8 e^{-\frac{\epsilon_{ion}}{k_B T}} \approx e^{20} e^{-\frac{\epsilon_{ion}}{k_B T}}$

(History back to decoupling has number of particles conserved).

This will ensure that there are enough high energy photons to ionize everything.

Population of photons from stars is a tiny fraction of the total photons in Cosmic Microwave Background.

7.4 Landmarks on the way back to the Big Bang

We want:

$$\begin{aligned}
\frac{\epsilon_{ion}}{kT} \approx & 20 \\
kT \approx & \frac{1}{20}\epsilon_{ion} \\
\frac{\epsilon_{ion}}{kT} \approx & 20 \\
T \approx & 4000K
\end{aligned}$$

So

$$\begin{aligned}
\frac{T_{DECOUPLING}}{T_{TODAY}} \approx & 1000 \\
\frac{a_{TODAY}}{a_{DECOUPLING}} \approx & 1000
\end{aligned}$$

Space expanded by a factor of 1000 since it was opaque. That is one landmark.

The second landmark is when the universe went from being radiation dominated to matter dominated.

There is a crossover between the two regimes. Let us see if we can estimate.

First, we need the ratios today. From (18) and (19):

$$\begin{aligned}
\rho_{matter} = & \frac{\rho_0}{a^3}, \text{ most of observable history} \\
\rho_{radiation} = & \frac{\rho_0}{a^4}. \text{ Let us take the energy density today} \\
\frac{\rho_{matter}}{\rho_{radiation}} \approx & 10^{-18} \times 10^{13} \\
\approx & 10^5, \text{ oops, we forgot dark matter, so} \\
\approx & 10^6
\end{aligned}$$

Using the fact that there are 10^8 photons per proton, energy of a photon is 10^{-4} eV, and a proton about 1 GeV.

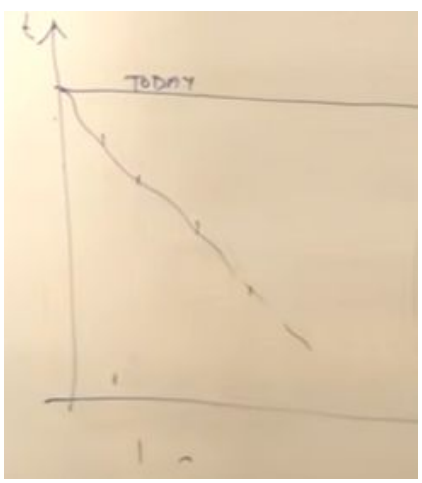
How did it behave in the past? In the past ρ_{matter} scales with 3 powers of the scale factor, and $\rho_{radiation}$ with 4. That means that $\frac{\rho_{matter}}{\rho_{radiation}}$ scales by 1 power of the scale factor. So back when $a(t) \approx 10^{-6} a_{today}$, $\rho_{matter} \approx \rho_{radiation}$: temperature was a million times greater, or a thousand times temperature of decoupling period. As we go further back, universe is more radiation dominated. We don't easily see back to this period.

The next important landmark was when the temperature was hot enough to create positrons.

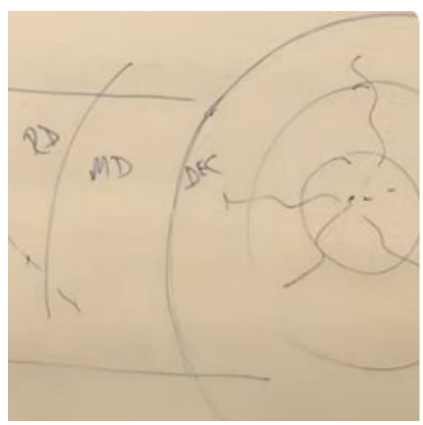
In flat space, only ratios of a matter. If $\kappa > 0$, we can make sense of $a(t)$ as the radius of the 3-sphere. If $\kappa < 0$, the universe started infinite.

Figure 25: The universe in a nutshell

(a) Looking backward along light rays, sampling every thousand light years or so.



(b) Looking back we see different time periods. The universe was optically opaque before decoupling. CMB comes from point where universe shifted from opaque. Neutrinos can come through from before. Gravitons. So, in principle, we may be able to look back further than decoupling.



When temperature gets to about $10^{14}K$, $\frac{a_T}{a(t)} \approx 10^{14}$, characteristic photon energy $\epsilon_{photon} = 0.5MeV$, the $E = mc^2$ of an electron. Two photons colliding can produce electron-positron pairs; also electrons and positrons can make photons. They can form a thermal equilibrium, and are no longer determined by "memory of the future". Thermal equilibrium at that temperature satisfies $N_{e^-} = N_{e^+} = N_\gamma$. Note that $N_{e^-} - N_{e^+}$ does not change, but $N_{e^-} + N_{e^+}$ does: it is bigger than today's value by 10^8.

What led to any difference?

8 Baryogenesis

8.1 The Question

Why is there so much entropy in the world?

- Why is the number of photons, which counts the entropy in the world[8], so much higher that the number of protons and electrons($10^8\times$ larger)?
- Why are the protons and electrons so few?

One explanation is the Sakharov conditions: there are three of them, and, if they are satisfied, there will be an imbalance of matter over antimatter.

Is this a legitimate question? Maybe the world started with a imbalance. But numbers focus your imagination. The excess is a fact, and maybe we could just assume it is that way.

In the very early universe there was a huge number of protons and a similar number of antiprotons (or quarks and antiquarks), a huge number of electrons and positrons, and a huge number of photons, all in thermal equilibrium. Then it cooled and the photons were left over, because they decoupled and the universe became transparent. The protons and antiprotons annihilated each other. The universe expanded slowly, so there was plenty of time for them to find each other. All that was left was the slight excess. So the question is not "why there were so many photons", but "why was there this slight excess of 10^{-8}"? We don't have a complete theory, but almost anything we write down always has a slight excess.

$$\begin{aligned} N_P + N_{e^-} - N_{\bar{P}} - N_{e^+} =& 0 \text{ This makes sense.} \\ \frac{N_P - N_{\bar{P}}}{N_\gamma} \approx& 10^{-8}, \text{ but why this?} \\ \frac{N_P - N_{\bar{P}}}{N_P + N_{\bar{P}}} \approx& 10^{-8} \end{aligned} \tag{34}$$

8.2 The Old Theory

Let us begin with a hypothesis, which is believed to be wrong. Let B denote the Baryon number, $\frac{1}{3}(N_q - N_{\bar{q}})$. The statement that there were more protons than antiprotons is also a statement about quarks and antiquarks. Let us suppose that Baryon number is like electric charge: it is conserved. Of course electric charge is the source of a long range force, but Baryon number is not. Protons create long range Coulomb forces, but neutrons don't. If Baryon number is truly conserved, then if there is an initial excess, there will always be the same excess, which will be frozen in. It appears to be highly conserved. Otherwise, what could a proton decay to?

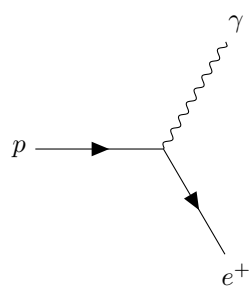
[8] Normally the entropy is the number of photons for black body radiation

- it must decay to something lighter;
- it must decay to something with a positive electrical charge.

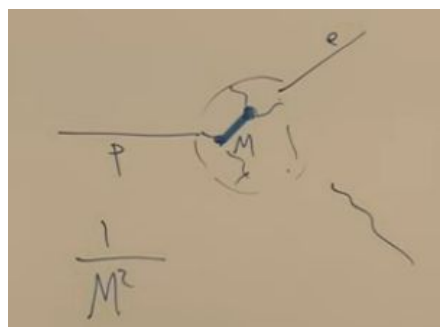
This leaves a positron and something electrically neutral, such as a photon (a neutrino is no good, as we'd have an even number of fermions). We don't know any deep fundamental reason why this can't happen.

Figure 26: Hypothetical proton decay. For some reason, that Standard Model does not permit the decay shown in Figure 26a, but there are extensions to the Standard Model where this could happen. We do know that the decay cannot be very fast, as the protons have been around for at least 13.7 billion years. Note that as protons decay, positrons are made.

(a) Hypothetical scenario where a proton decays into a positron and a photon.



(b) Somewhere in the guts of the Feynman diagram, assume there is a very, very heavy particle, outside the Standard Model, say $10^{16} GeV$, then the process is very slow, and is suppressed by powers of the heavy mass.



For some reason, that Standard Model does not permit the decay in Figure 26a, but there are extensions to the Standard Model where this could happen. We do know that the decay cannot be very fast, as the protons have been around for at least 13.7 billion years. So we think that the excess has been built in from the very beginning.

There is another symmetry of nature that isn't really a symmetry. Sakharov wrote a couple of years after it was discovered that it wasn't a symmetry: particle-antiparticle reflection symmetry. This says that if particles are replaced with reflection we can't tell the difference.

Table 2: Until the 1960s it was believed that the following 3 symmetries existed

	Name	Description
C	Charge Conjugation	If positive charges are replaced with negative, and vice versa, we can't tell the difference
P	Parity	If particles are replaced with reflection we can't tell the difference
T	Time	If the clock is run backwards we can't tell the difference

We now know that the symmetry is CPT (In the Scrödinger equation, $t \to -t, \psi \to \psi^*, i \to -i$).

$$
\begin{aligned}
i\hbar\frac{\partial\psi}{\partial t} =& \big[-\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + V(x)\big]\psi \\
-i\hbar\frac{\partial\psi^*}{\partial(-t)} =& \big[-\frac{\hbar}{2m}\frac{\partial^2}{\partial(-x)^2} + V(-x)\big]\psi^* \text{ the same equation if } V(x) = V(-x)
\end{aligned}
$$

Imagine (before Sakharov) that CP was a symmetry: then you might be able to swap particles and antiparticles and reflect left & right. But if there were a complete symmetry, why the imbalance? It doesn't sound right.

8.3 The Sakharov Conditions

Sakharov showed that the following three conditions were sufficient to account for the observed excess of protons over antiprotons.

1. Baryon number violation–Section 8.3.1
2. C-symmetry and CP-symmetry violation–Section 8.3.2
3. The universe is expanding fast enough to break T symmetry–Section 8.3.3

8.3.1 Baryon number violation

This begins with the idea that there *was* a balance: that there was no bias towards particles or antiparticles. The only way to get unbalanced is if conservation of Baryon number is not correct. If something like Figure 26a can happen, the number of protons can change. The first requirement for a theory that starts with balance is to have a mechanism that violates conservation. That was the first of the Sakharov conditions.

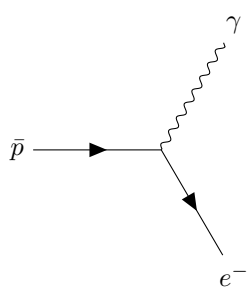
If Baryon number conservation is not a good symmetry, it must be very weakly broken, and have a very small probability per unit time, since protons are very, very old. Every known unified theory violates Baryon number conservation. You might ask why, in the known theories, the proton is so stable. Somewhere in the guts of the Feynman diagram, Figure 26b, assume there is a very, very heavy particle, outside the Standard Model, say $10^{16}GeV$, then the

process is very unlikely, and its amplitude is suppressed by powers of the heavy mass.

If the proton has extra energy, the factor of $\frac{1}{M^2}$ in Figure 26b becomes $\frac{1}{(M-E)^2}$, so there will be less suppression ($Energy = mc^2 + E$). Normally E is negligible. Theories of this type can explain stability.

All current theories include the *possibility* of Baryon violation. The universe was very hot at beginning! But Baryon violation by itself isn't enough by itself to give an average excess of protons over antiprotons: for every event like Figure 26a, which reduces baryon number, there is a charge conjugated process that increases baryon number–Figure 27.

Figure 27: Decay of an antiproton



If we believe in particle-antiparticle symmetry, on average there will be as many Figure 27 decays as Figure 26a. The process could decay antiprotons, or run in reverse. If we started with an equal population of quarks and antiquarks, and try to rely only on violation of Baryon number, it would not be an efficient way to create an excess. Moreover, in the very early universe there were lots of electrons, positrons and photons, so it was possible for the reactions to go backwards. There is a statistical process going on, so it is only statistical that there would be as many protons as antiprotons. The excess of protons over antiprotons cannot be a statistical effect.

It is quite true that the excess is a small number, 10^{-8}, but how many protons are there in the observable universe? About 10^{80}. How big would you expect it to be if it were a pure statistical effect? $\sqrt{N}$ is the expected fluctuation in a variable. In a world with 10^{80} protons, we'd expect a fluctuation of $\sqrt{10^{80}} = \sqrt{10^{40}}$. If excess were around this size it might be statistical.

If it were statistical fluctuation, it would be hard to understand why it is the same everywhere. If there were anti-galaxies, we'd expect high energy cosmic rays to have as many anti-nuclei as nuclei[9]. We don't see this. (We do see antiprotons, however, but they can be explained by collisions in the upper atmosphere).

[9] LS stated that Helium nuclei have been seen in cosmic rays.

If our galaxy were matter, and the Andromeda galaxy were antimatter, we'd expect to see a lot of annihilation in between.

8.3.2 C and CP violation

Particle-antiparticle symmetry *suggests* that the universe was created symmetrically, although it doesn't prove it. In order to account for the lack of symmetry between particles and antiparticles, this charge conjugation symmetry must fail, as must CP symmetry. Baryon violation isn't enough: we need something to push it in a particular direction. We need bias. We need violations of Particle-Antiparticle symmetry, and both C and CP symmetry.

Is there evidence of violation of particle-antiparticle symmetry? Examples are difficult to find, but they do exist; we have experimental evidence that particles and antiparticles don't behave the same way. The simplest example is the B meson, which is made of a bottom quark and an anti-down quark: it has an anti-particle.

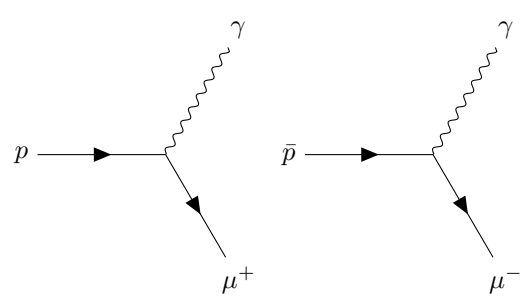
- $B \to K^+ + \pi^-$
- $\bar{B} \to K^- + \pi^+$

The rates for these to happen are measurable, and *they are different*. One is about $\frac{2}{3}$ more important than the other, which is a fairly gross example of violation of particle-antiparticle "symmetry".

There are fundamental processes buried deep in Feynman diagrams that are unbalanced between particles and antiparticles. It is not the case that the processes depicted in Figures 27 and 26a have to have equal probabilities.

There is a theorem of relativistic field theory that the total half life of the proton and antiproton have to be the same[10], but Figures 27 and 26a can have different half lives; there is more than one way a proton can decay. For example a proton can also decay into a μ and a photon–Figure 28

Figure 28: A proton can also decay into a μ and a photon



[10] https://youtu.be/gdFldkitkJA?t=2991

The theorem says we calculate all the ways the proton can decay, and it must give the same rate as the total for the antiproton. It does not say that the rates for the individual diagrams should be the same: it doesn't say that any particular decay has to have a symmetry.

If, in particular, at some level charge conjugation symmetry is violated, two processes, e.g. Figures 27 and 26a, need not be the same. Somewhere in the laws of physics there is a bias towards matter. We need to add this to baryon number violation to give it a push[11].

Question: if the total decay rates are the same, why don't we have equal numbers?

Because the μ particle is heavier than an electron, so we could be in situation where it hot enough...I need to explain why this can happen. It has to do with the environment.

How do we overcome the fact that the proton is so incredibly stable? 10^{34} years, compared with the age of the universe, 10^{10} years. On that scale the proton is very stable, so why don't we get to ignore this kind of thing?

- The universe, in its early stages, was very, very hot
- so protons or quarks were constantly engaged in very high energy collisions.
- Protons had very high energy
- at some point the everage kinetic energy is high enough that $\frac{1}{(M-E)^2}$ does not suppress baryon violation
- Protons are normally long lived, but when one is heated to a high enough temperature, the excess mass in Figure 26b is not longer important

In short, in the early universe, when temperature are very high, these processes can happen.

The statement that the total decay rate for protons and antiprotons is the same is a statement about zero temperature: it is a statement about a proton at rest in an environment where it has no excess energy. If the proton is being kicked around, decay rates don't have to be the same.

8.3.3 Universe is expanding enough to break T symmetry

We still have a problem with CPT symmetry. In all known theories that have relativity and quantum mechanics build in, we have CPT symmetry: replace each particle with its antiparticle, reflect in a mirror, then run film backward. So, in thermal equilibrium, if the universe was just not changing, in thermal equilibrium forward time is the same as backward time. In such a universe, if we had time reversal symmetry, CPT symmetry would give us CP symmetry. If there is no bias to time, we are back with no excess of protons.

[11] Sakharov wrote down his conditions within a year or two of the discovery of CP violation

But the universe was not in thermal equilibrium: it was expanding *fast*, fast enough that it could not be considered in equilibrium. In a rapidly expanding phase of the universe, we don't have to worry about time symmetry. If time symmetry is broken by the expansion of the universe, then we are in business. Universe must be expanding fast enough that these microscopic processes don't have time to adjust themselves to the equilibrium configuration. Expansion is suddenly slowed.

These appear to be true, and they are sufficient to explain fact of excess: it isn't a surprise. But nobody knows enough to make a calculation of magnitude of imbalance in Equation (34)–10^{-8}

9 Inflation

(The first two sections were presented at the end of the previous lecture.)

9.1 The Question

The inflationary universe idea was put forward to explain why the universe is so homogeneous and isotropic. This became important when the Cosmic Microwave Background was discovered, and the CMB quickly became a high precision thing. The temperature was measured to be $2.7K$, and was the same in every direction. The rough idea that the universe was isotropic became a high precision thing. Why is it so isotropic? It must have become this way very early, as lumpiness tends to increase with time, because of gravity. The universe must have started very early–before galaxies, at the decoupling time– being homogeneous and isotropic. But not absolutely homogeneous and isotropic, since there are galaxies and clusters. The lumpiness must have been small. Gravity is different from other theories: it increases lumpiness, instead of letting it diffuse.

Cosmologists have been able to estimate the inhomogeneity that was necessary at the time of decoupling for galaxies to have formed later. They characterized inhomogeneity as $\frac{\delta\rho}{\rho} \approx 10^-5$, the mean excess density in a lump. Now we had a number, which we could try to explain. We still don't know why this is the correct value. The existence of this number focused attention on the questions. First, why is it so homogeneous? Later, why is is a bit inhomogeneous?

The explanation from Alan Guth in 1980 was that the universe expanded very fast, and stretched. Stretching an inhomogeneous universe makes it homogeneous on the scales that are relevant. The universe inflated–expanded exponentially– and stretched itself so much that it smoothed out the inhomogeneities. During the expansion period, temperature wasn't terribly relevant, it was the rapid expansion.

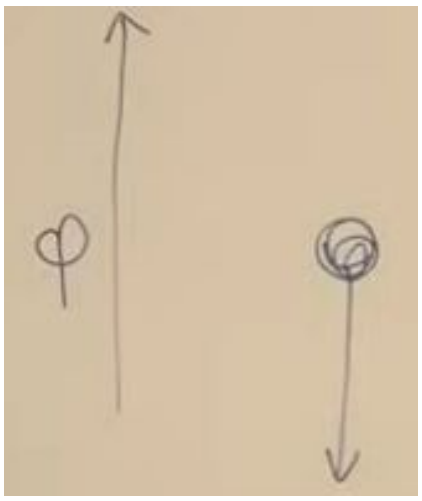
9.2 The Inflaton Field

9.2.1 End of Lecture 8

This means viscosity: consider a stone falling through honey–Figure 29a.

Figure 29: Viscosity example–Falling stone

(a) The height is ϕ, and stone is subject to a force arising from potential energy $V(\phi)$. Assume force is upward, so V increases in downward direction.



(b) If $V(\phi)$ is a rather shallow hill, and the viscosity coefficient is large, the stone will take a long time to roll down the hill.

Then the stone satisfies the equation of motion (35).

$$\begin{aligned} F &= -\frac{dV}{d\phi} \\ \ddot{\phi} &= -\gamma\dot{\phi} - \frac{dV}{d\phi}, \text{ Newton with } m = 1 \text{ and viscosity} \\ \dot{\phi} &= \frac{F}{\gamma}, \text{ at terminal velocity} \end{aligned} \tag{35}$$

If $V(\phi)$ is a rather shallow hill–Figure 29b–and the viscosity coefficient is large, the stone will take a long time to roll down the hill.

We want to make ϕ the value of a field, and we will want to be in a situation where the field evolves very slowly because of a lot of friction, which will drive inflation. We assume that the universe is filled by a scalar field, the inflaton field ϕ, which is pretty much uniform in space. Think about the energy stored in the field. Follow the field in an expanding box, size $(a(t))^3$. The total energy,

kinetic plus potential, is given by the following equation.

$$E = a^3(t)\big[\frac{\dot{\phi}^2}{2} + V(\phi)\big], \text{ no spatial gradient since uniform}$$

(We have assumed that the field doesn't vary much in space. Because space expands, the variation in the field has become stretched out, so much that it doesn't vary across a box.)

We use Lagrange's equation to derive the equation of motion. First calculate the Lagrangian.

$$\begin{aligned}
\mathscr{L} =& a^3(t)\big[\frac{\dot{\phi}^2}{2} - V(\phi)\big] \text{ and substitute in Lagrange's equation} \\
\frac{d}{dt}\frac{\partial \mathscr{L}}{\partial \dot{\phi}} =& \frac{\partial \mathscr{L}}{\partial \phi}. \text{ The first term is} \\
\frac{d}{dt}\frac{\partial \mathscr{L}}{\partial \dot{\phi}} =& \frac{d}{dt}\big[\dot{\phi} a^3(t)\big], \text{ whence Lagrange's equation becomes:} \\
\frac{d}{dt}\big[\dot{\phi} a^3(t)\big] =& -a^3(t)\frac{dV}{d\phi}. \text{ Expanding the time derivative we get:} \\
a^3\ddot{\phi} + \dot{\phi}\big(3a^2\dot{a}\big) =& a^3 F. \text{ Now divide by } a \\
\ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} =& F, \text{ and use Equation (2)} \\
\ddot{\phi} + 3H\dot{\phi} =& F \qquad (36)
\end{aligned}$$

9.2.2 Lecture 9 begins

We postulate a hypothetical, undiscovered scalar field. Fields have energy, which depends on the value of the field. Let us hypothesize an energy, Figure 30.

Figure 30: The energy density that would be stored in space by our hypothetical, undiscovered scalar field. We haven't shown zero energy for $\phi = 0$ because we don't know where its zero is. We call this "potential energy" as it is not associated with time dependence of the field.

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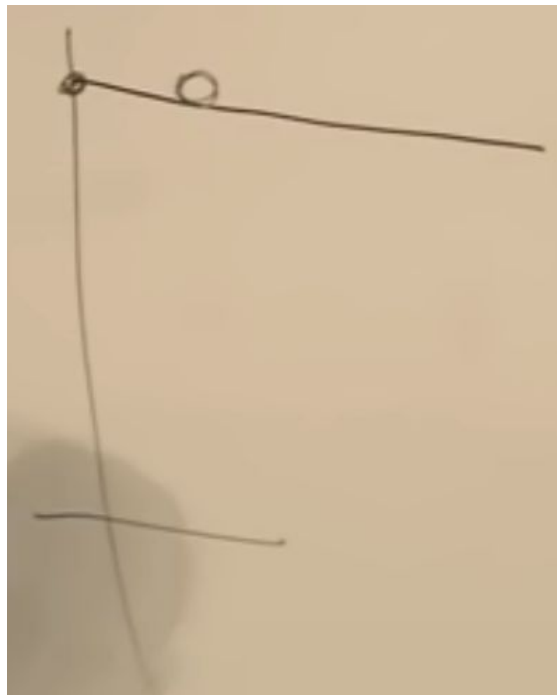


A scalar field has kinetic energy $\frac{\dot{\phi}^2}{2}$, so the field energy in a small unit box is $\frac{\dot{\phi}^2}{2} + V(\phi)$, which is isomorphic to the motion of a particle in a field ϕ.

(I have skipped over the remainder of the review of Lecture 8)

We resume with Equation (36), which should be compared with (35) and Figure 29a. Note the "cosmic friction" term $3H\dot{\phi}$. Rock accelerates, but friction slows it.

Suppose that the viscosity does not depend on time. In the beginning there is no friction, and the velocity increases; viscosity plays a bigger role, until eventually the stone reaches (really approaches) terminal velocity$-\frac{F}{3H}$. At his point kinetic energy is negligible, compared to potential energy.

What happened to energy conservation? Energy conservation is a consequence of time translation invariance, and $a(t)$ introduces a dependency on time.

9.3 How does ϕ affect the expansion?

This is one half of the story for expansion in the presence of a scalar field. But how does ϕ affect the expansion? For simplicity we will confine ourselves to the flat space case. The Friedmann equation (7) becomes:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \underbrace{\left[\frac{\dot{\phi}^2 + V(\phi)}{2}\right]}_{\rho}$$

Let us assume that V is large and its slope is gradual, as depicted in Figure 30. Since the slope is gradual, the force is small.

(LS emphasized that the inflaton field, ϕ is hypothetical. Other scalar field wouldn't do the job, especially the Higgs field.)

In Figure 30 the "ball" slowly creeps down the hill.

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 \approx \frac{8\pi G}{3} V(\phi)$$

Now if H^2 is approximately constant, because the right hand side is approximately constant, we have:

$$\begin{aligned} \frac{\dot{a}}{a} &= H \\ \dot{a} &= Ha \\ a &\propto e^{Ht}, \text{ and} \\ H &= \sqrt{\frac{8\pi G}{3} V(\phi)} \end{aligned} \tag{37}$$

This is a theory of the very early universe, when it was expanding at a large rate. Cosmologists believe that the doubling time was $10^{-32}s$. This is not a theory of the current universe.

Question about how gradual variation can cause exponential growth.

- $V(\phi)$ is large, and it drives expansion'
- the gradient of $V(\phi)$, and it is driving ϕ.

The viscous drag affects ϕ, not a.

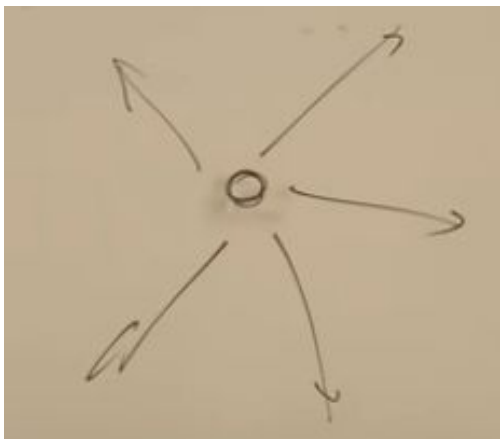
Why should we be interested in a theory in which the universe was expanding very rapidly? Say by a factor of e^{60}. There were two puzzles.

- They wanted to explain why magnetic monopoles were undiscovered.
- Homogeneity at the time of decoupling, from CMB.

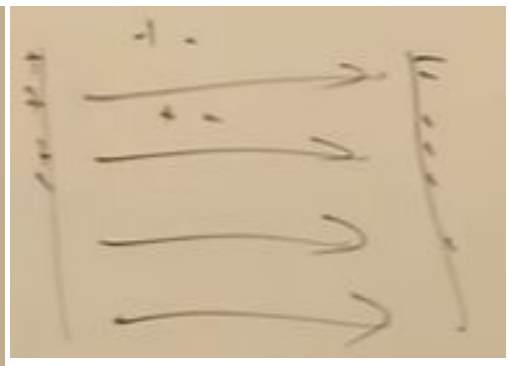
9.3.1 They wanted to explain why magnetic monopoles were undiscovered

Figure 31: A magnetic monopole radiating like a Coulomb field

(a) A magnetic monopole is a hypothetical particle with a magnetic field radiating out like a Coulomb field



(b) Electric field collapsing: the particles in the middle migrate to the plate that has opposite charge.



According to the current theory, they have never been discovered in the laboratory because they are very heavy–near the Planck mass, say $10^{19} proton$. The early universe was *hot*. Far enough back photons could scatter and produce electron-positron pairs. Further back there was enough energy to produce proton-antiproton pairs. Even further back it could have created magnetic monopoles. Where wee they?

Magnetic monopoles don't decay easily: protons and electrons and things don't have a magnetic charge. We can work out how likely they are to annihilate each other, but that isn't likely. We would have expected to find them, and maybe not a lot, since the universe has expanded by a factor of 1000 since decoupling. They'd have an interesting effect on the magnetic fields of galaxies. If you have charged particles in an electric field, they will discharge it–Figure 31b. Similarly if there was a large population of magnetic monopoles, and they lined near galaxies, then, over the life of the universe they would have discharged the magnetic field. But this hasn't happened.

So there is no significant population of magnetic monopoles, even though the standard theory predicts they should have been created, and would not dilute them enough to make them undetectable.

9.3.2 Homogeneity at the time of decoupling

The other set of facts that need to be explained is Homogeneity at the time of decoupling. The flux and temperature of the CMB were measured in different

directions with very high precision, 1 part on 10^4 in 1980, and it was absolutely uniform. Why was everything so smooth at the time of decoupling? A small universe, very, very hot, would be expected to have all sorts of fluctuations, and a standard expansion would not have flattened it to the degree that had been observed.

Alan Guth guessed that very early, after the monopoles had been produced, and after the universe had a very complicated texture, an enormous expansion came, it diluted the monopoles, so there were many fewer than had been estimated, and it also diluted the bumps. Nobody can prove this: maybe the universe just started smooth and stayed that way; maybe it started with no monopoles, but there was no good reason for it.

A very rapid expansion can solve both problems.

It was estimated that, in order to make the universe the way it is, from a known starting point, or an unknown starting point that isn't smooth, you need at least 60 e-foldings, i.e multiplication by e^{60} to get to the known smoothness. So Ht needs to be *at least* 60, which means that the ramp in Figure 30 must have been rather shallow.

That is where the theory was around 1980.

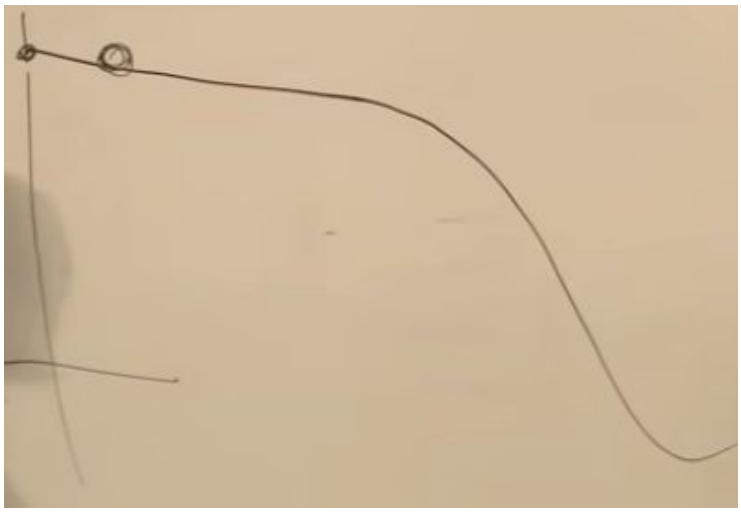
9.4 The current theory

If the number of e-foldings were exactly 60, we'd be on the edge of seeing what the inflation was invented to get rid of. If so we should be on the threshold of seeing some additional curvature in the universe. The Planck satellite will search for curvature.

Expansion cools things so, in a classical universe inflation should have cooled things down.

Why didn't it go on indefinitely? We are here, so we know it didn't. Sop something must have happened. Figure 32 depicted the most likely guess: the potential came to the end of a plateau.

Figure 32: The potential came to the end of a plateau. We don't know why, just that this is a good representation of cosmological data.



The bottom of the curve represents today: the potential energy is very small. This is the model that works.

- How did the universe get to the top left in the first place? Nobody knows: it is part of the initial condition. We know from CMB data that there was a period of inflation, so we must have started at the top left, then gradually rolled down the hill, taking long enough that 60 e-foldings took place.

- What happens when we get to the end of the plateau? Viscosity became less important, for two reasons.

  - As you go down the hill, H gets smaller. From (37) $H \propto \sqrt{V}$, so H gets smaller as V gets smaller.
  - F get bigger as we go down the hill.

  Viscosity becomes less compared to F, the ball would slide rapidly down the hill, and inflation would stop.

- In the process of sliding down the hill, energy was released (Possibly inflatons were emitted, which decayed into other particles). It is true that in an expanding universe energy is not strictly conserved, but most of the potential energy would be converted into something else, the velocity of the field initially. When we get to the bottom, where is the energy?

- The energy was converted to heat ("reheating"), and to particles: that is what we call the Big Bang, where standard cosmology starts. It is essential that the temperature be too low to create monopoles.

- What about the potential energy at the bottom? This is known as dark energy, or vacuum state energy.

We have a unified theory of inflation and dark energy.

Why does the potential have the shape depicted in Figure 32? LS suggests that the explanation is anthropic. A large part of Figure 32 arises because we need it.

10 Inhomogeneities and quantum fluctuations

10.1 Question (End of Lecture 9)

Why isn't the universe completely smooth? The amount of inflation would have smoothed the universe enough to eliminate the structure we have. We are not talking about galaxies, nor clusters, but superclusters of superclusters. They form a structure that fills the universe; it is not completely homogeneous in those scales. It is largely understood, but mainly through simulations.

The simulations start by assuming that the universe was not completely smooth, and it started with a certain amount of structure called the "fluctuation spectrum". They allow gravity and motion of objects to evolve, and see that it produces structure.

On bigger scales, gravity is not that important. On the scale of a galaxy, or a star, or a planet, gravity is what pulls things together. On bigger scales, there is structure that is largely not coming from gravity; it would be there even if there were no gravity. What it takes is to start with some inhomogeneity.

LS gave some images to the students that showed the Large scale structure of the Universe.

- Figure 33a is a photograph of the sky, but deep, deep astronomy: billions of light years out, high redshifts. It has some structure, and is not featureless: it has a filamentary structure. The filaments aren't single galaxies, they are millions of galaxies. The filaments intersect and form a complicated web, rather like Figure 33b.

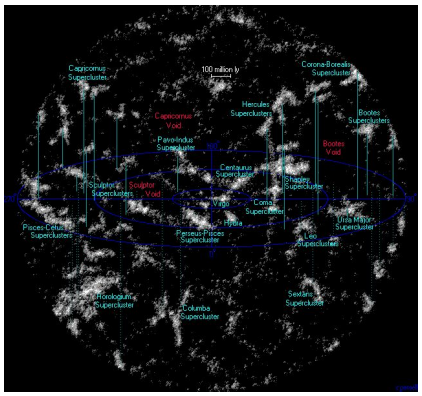
- Figure 33c is a simulation. It started with a big bunch of particles, with some spectrum of fluctuations, a little randomness in the velocity distribution. Let it evolve for a while. Maybe start with a uniform distribution of matter, and a little variation in velocity.

- Figure 33d shows how light forms patterns at the bottom of a pool of water.

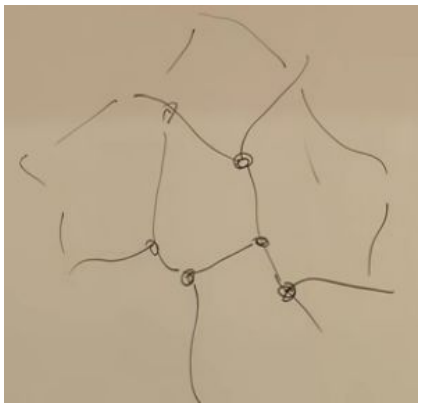
Figures 33a, 33c, and 33d have a remarkable similarity, which can be explained by caustics.

Figure 33: LS gave some images to the students that showed the Large scale structure of the Universe. Figures a, c, and d are images that I have found that appear to tell the same story.

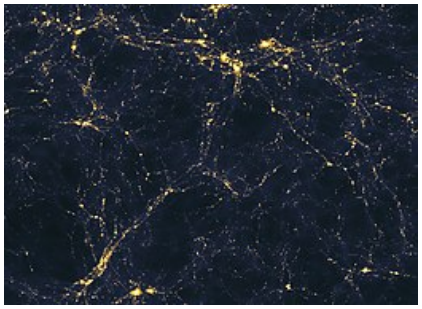
(a) The sky, but deep, deep astronomy: billions of light years out, high redshifts. It has seem structure, and is not featureless. It has a filamentary structure. [5]



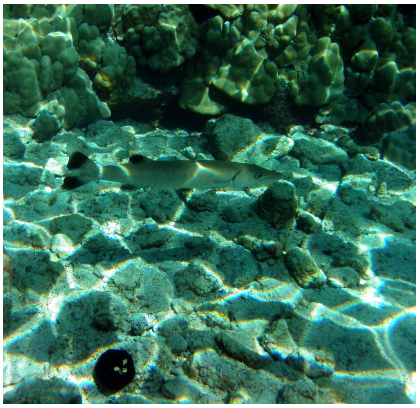
(b) A simple representation. At the points of intersection things are particularly bright: they are not single galaxies. There are big voids between the filaments.



(c) Simulation. Start with a big bunch of particles, with some spectrum of fluctuations, a little randomness in the velocity distribution.[6]



(d) Light forms patterns at the bottom of a pool of water[7]

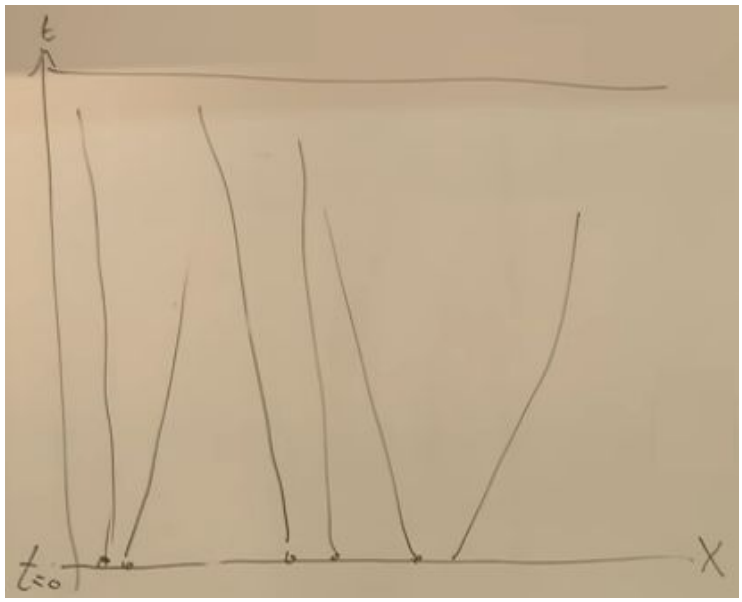


10.2 Caustics

If we were dealing with only one dimension, caustics would be bright spots. In two dimensions they are replaced by lines of high density.

In Figure 34 we have some structure imposed by waviness in the surface of the water. When light rays come in they are refracted a little, and the light goes off at an angle that varies from place to place. In the one dimensional case there are spots of high intensity.

Figure 34: Large structure of universe, with time running upwards *or*, swimming pool with depth t running upwards. In the latter case, the top of the pool is at the bottom of the picture; rays of light come in at slightly different angles, because of waves on the surface, then hit the top of the picture (bottom of the pool).



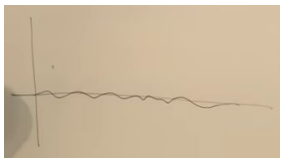
We can think of Figure 34 as waves, or as particles arriving with different velocities at $t = 0$. Let us assume that the particles are uniformly distributed when they start, with no tendency to bunch up. We define n as the number of particles along the $t = 0$ axis, so $\frac{dn}{dx}$ is the density.

$$\begin{aligned}
\frac{dn}{dx} =& 1. \text{ Since it is uniform, set it to 1.} \\
y(t) =& x + v(x)t, \text{ where } y(t) \text{ denotes the position after time } t \\
\frac{dy}{dx} =& 1 + v'(x)t \\
\frac{dn}{dy} =& \frac{dn}{dx}\frac{dx}{dy}, \text{ the density after time } t \\
=& \frac{1}{1 + v'(x)t}
\end{aligned}$$

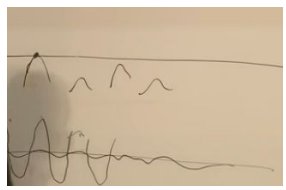
The bright spots are where $1 + v'(x)t = 0$.

Figure 35: When does $1 + v'(x)t = 0$?

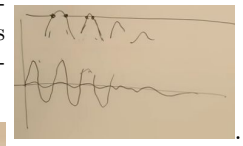
(a) $-v'(x)t$ (This is meant to look random rather than periodic.)



(b) As t increases we get a point where $v'(x)t = -1$, so we have a bright spot, or a big flux of particles. The rest of the peaks don't cross the line, however.



(c) As t continues to increase we get multiple bright spots.



.

Now, if we do this in two dimensions you get something that looks like Figure 33b. Hot spots intersect. In 3 dimensions you get surfaces, intersecting other surfaces, etc. This is what you'd expect in a world without gravity.

Where did the variations in velocity come from?

10.3 Digression, leading to Planck Units

We do know a highly accurate description of elementary particles. It has a lot of parameters. Depending on the version that turns out to be true, the number of parameters could vary from 25 to 150: masses of quarks, masses of electrons, masses of muons, the coupling constants, the value of they. electrical charge, and a list of what the particles are. Some of the particles are related to each other, others are not. So there is a very accurate theory that describes everything that is known about elementary particles. Thousands of experiments. In principle it can explain nuclei, atoms. Quantum Electrodynamics is part of it, so it can explain molecules, chemistry.

We also have a classical theory of gravity, General Relativity. Again it is highly accurate theory. Does it have any parameters? There is Newton's constant of gravitation. There are masses of particles. These aren't thought to be deeply fundamental. There are three parameters, however, that are universal.

1. The speed of light. Einstein's principle, was that *all* particles, signals, etc moves slower than the speed of light.

2. Planck's constant: the uncertainty principle applies to *all* measurements.

3. Newton's constant applies to *all* objects.

The use of the word *all* is really deep. Planck realized that there were 3 fundamental units, and 3 fundamental constants, so he could define basic units that make the 3 constants unity.

$$
\begin{aligned}
[c] =& \frac{[L]}{[T]} \\
[\hbar] =& \frac{[M][L^2]}{[T]} \\
[G] =& \frac{[L^3]}{[M][T^2]}
\end{aligned}
$$

We define new units:

$$
\begin{aligned}
[L] =& c^a \hbar^b G^d \\
=& \frac{[L^a]}{[T^a]} \frac{[M^b][L^{2b}]}{[T^b]} \frac{[L^{3d}]}{[M^d][T^{2d}]} \\
=& \frac{[L^a]}{[T^a]} \frac{\cancel{[M^b]}[L^{2b}]}{[T^b]} \frac{[L^{3b}]}{\cancel{[M^b]}[T^{2b}]}, \text{ if } d = b \\
=& \frac{[L^{a+5b}]}{[T^{a+3b}]} \text{so we need:} \\
a + 5b =& 1 \\
a + 3b =& 0 \\
2b =& 1 \\
b =& \frac{1}{2} a = -\frac{3}{2} \\
[L] =& \sqrt{\frac{\hbar G}{c^3}} \\
\approx& 10^{-35} \text{ metres.}
\end{aligned}
$$

$$
\begin{aligned}
[T] =& \frac{[L]}{c} \\
=& \sqrt{\frac{\hbar G}{c^5}} \\
\approx& 10^{-43} \text{ sec.}
\end{aligned}
$$

$$
\begin{aligned}
[M][c^2][T] =& [\hbar] \\
[M] =& \frac{\hbar}{c^2[T]} \\
=& \frac{\hbar}{c^2}\sqrt{\frac{c^5}{\hbar G}} \\
=& \sqrt{\frac{\hbar c}{G}} \\
\approx& 10^{-8} \text{ Kg–about a dust mote}
\end{aligned}
$$

At the time the lecture ws given the LHC was probing down to a distance of 10^{17} Planck lengths: we don't know the phenomena below this scale. The Standard Model works well for the LHC level.

We don't know why the parameters are the values they are, or why they appear to be so fine tuned.

Why are fundamental particles so much lighter than the Planck mass? By a factor of 10^{-25}. This is connected with the lightness of the Higgs Boson.

The Planck mass is the mass of the smallest possible black hole: it carries one bit of information.

10.4 Inhomogeneities at the time of decoupling

Figure 36 shows a map of the universe–the Wilkinson Microwave Anisotropy Probe (WMAP)–which depicts the thermal remnant of radiation, coming from the era of decoupling. Figure 37 depicts concentric shells, centered on us, and ending when the universe was opaque. The radiation is redshifted by a factor of 1000: ordinary light is redshifted to 2K blackbody radiation. The radiation is isotropic and ariound 2.7K to high precision. Perfect isotropy would also mean perfect homogeneity, but the universe is not perfectly homogeneous. So there must have been some very small inhomogeneity.

Figure 36: Map of the Universe[8]. The galactic plane is a horizontal band in the middle of the image, which has been subtracted out. To a first approximation it is very isotropic.

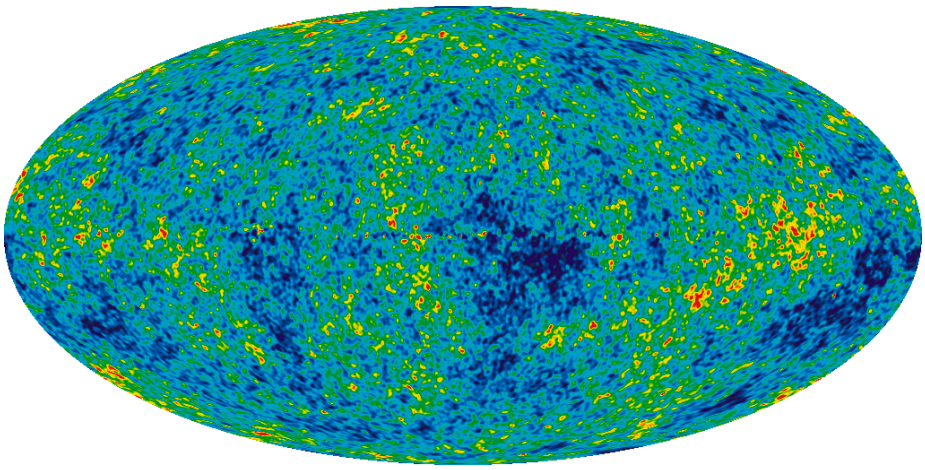


Figure 37: Concentric shells, centered on us, and ending when the universe was opaque

Figure 36 shows the temperature distribution, and $\frac{\delta T}{T} \approx 10{-}5$. That number comes from two different places:

- simulations, tracing back the inhomogeneities that must have been present to get our present universe;
- direct observation, which shows structure on all scales. There are features at all scales.

Where do temperature fluctuations come from? From mass fluctuations, so $\frac{\delta\rho}{\rho} \approx 10^{-5}$ at decoupling. We need a theory to explain this variation. There is a pattern: scale invariance.

The current theory is that they originated from quantum fluctuations. It depends on two things:

- quantum fluctuations, quantum zero-point motion of the harmonic oscillator, the damped harmonic oscillator;
- the equations of a scalar field in the expanding universe.

10.4.1 The undamped harmonic oscillator

We start with the undamped oscillator:

$$\begin{aligned} E =& \frac{\dot{X}^2}{2} + \frac{\omega^2 X^2}{2}, \text{ Hamiltonian} \\ E =& (n + \frac{1}{2})\hbar\omega, \text{ quantum mechanically} \end{aligned}$$

Average kinetic and potential energy are the same.

- n counts excitation;
- $\frac{1}{2}\hbar$ is the zero-point energy. It comes from the uncertainty principle.

In the ground state, on average:

$$\begin{aligned} \omega^2 X^2 =& \frac{1}{2}\hbar\omega \\ X =& \sqrt{\frac{\hbar}{2\omega}} \\ X(t) =& \sqrt{\frac{\hbar}{2\omega}} e^{\omega t} \\ \dot{X(t)} =& \sqrt{\frac{\hbar\omega}{2}} e^{\omega t} \end{aligned}$$

That is the theory of zero-point energy. No matter how much you try to cool the system, zero-point energy is still there.

10.4.2 The damped harmonic oscillator

The equation of motion for the *damped* harmonic oscillator is:

$$\ddot{X} + \gamma\dot{X} + \omega^2 X = 0 \tag{38}$$

$$\text{Try } X = e^{\alpha t} \text{ as a solution } \alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega^2}}{2}$$

We have four cases.

1. under-damped;
2. over-damped;
3. critically damped;
4. no restoring force.

$$\begin{aligned} &\text{If } \gamma^2 < 4\omega^2 \\ &X \propto e^{-\gamma t} e^{\pm\sqrt{4\omega^2 - \gamma^2}} \end{aligned}$$

This is under-damped: the particle oscillates, but there is an exponential decay to the amplitude. In the over-damped region:

$$\begin{aligned} &\text{If } \gamma^2 > 4\omega^2 \\ &\alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega^2}}{2} \end{aligned}$$

Both α are negative, so the solutions decay with time.
There is also a transition between the two cases, the critical case.

$$\begin{aligned} \text{If } \gamma &= 2\omega \text{ the two solution coincide} \\ X &= e^{-\gamma t} \text{ is one solution. Let us try} \\ X &= te^{-\gamma t} \text{ as a second independent solution to (38)} \\ \dot{X} &= (1 - \gamma t)e^{-\gamma t} \\ \ddot{X} &= \gamma(\gamma t - 2)e^{-\gamma t}, \text{ then (38) becomes:} \\ \gamma(\cancel{\gamma t} - 2) + \gamma(1 - \cancel{\gamma t}) + \omega^2 &= 0 \\ \gamma &= \omega^2 \end{aligned}$$

No restoring force,

$$\begin{aligned}
\text{If } \omega =& 0 \\
\alpha^2 + \gamma\alpha =& 0 \\
\alpha(\alpha + \gamma) =& 0, \text{ whence} \\
\alpha =& 0, \text{ or} \\
\alpha =& -\gamma \\
X =& c_0 + c_1 e^{-\gamma t}
\end{aligned} \tag{39}$$

Particle asymptotically comes to rest.

Imagine the following scenario: ω is not actually a constant; it varies slowly. It starts large, in the oscillating regime. Now γ gets larger, and the frequency decreases–Figure 38. That is the system we will be interested in.

Figure 38: Varying ω. It starts large, in the oscillating regime. Now γ gets larger, and the frequency decreases. Eventually particles comes to rest as in Equation (39)

In order the see the relevance of this, consider the equation of motion of the Inflaton field (35).

Figure 39: Potential for inflaton field. We thought about a potential that is very flat for some energy density, then a cliff. We don't understand why the potential has this behaviour, but we know from observation that it has to be this way for things to work out. The "ball" rolls slowly down falls of the cliff, and goes to the bottom.

Let us make an approximation, and suppose that the flattish part is really flat, so $V' = 0$. The the field satisfies the wave equation.

$$\frac{\partial^2\phi}{\partial t^2} - \frac{\partial^2\phi}{\partial x^2} - \frac{\partial^2\phi}{\partial y^2} - \frac{\partial^2\phi}{\partial z^2} = 0, \text{ if not flat } -\frac{dV}{d\phi} \tag{40}$$

Now, in an expanding universe:

$$\frac{\partial^2\phi}{\partial t^2} + \underbrace{3H\frac{\partial\phi}{\partial t}}_{\text{zero if not expanding}} = 0 \text{ if } \phi \text{ does not vary from point to point}$$

There has to be another term if ϕ does vary from place to place: the term has to be such that we will reproduce (40) if a is constant. We might guess:

$$\frac{\partial^2\phi}{\partial t^2} + 3H\frac{\partial\phi}{\partial t} - \frac{\partial^2\phi}{\partial x^2} - \frac{\partial^2\phi}{\partial y^2} - \frac{\partial^2\phi}{\partial z^2} = 0$$

This might be correct of we were measuring in standard units, such a light seconds, but we are not[12]. We need to incorporate the scale factor. Suppressing y and z for clarity:

$$\frac{\partial^2\phi}{\partial t^2} + 3H\frac{\partial\phi}{\partial t} - \frac{1}{a^2}\frac{\partial^2\phi}{\partial x^2} = 0 \tag{41}$$

Let us try:

$$\phi(\vec{x}, t) = \phi_k(t)e^{i\vec{k}\cdot\vec{x}} \text{ in (41)}$$

$$\ddot{\phi_k}(t) + \underbrace{3H}_{\gamma}\dot{\phi_k}(t) + \underbrace{\left(\frac{k}{a(t)}\right)^2}_{\omega^2}\phi_k(t) = 0$$

[12] https://youtu.be/NZ_CvI9Ying?t=4915

Figure 40: We want to solve for a function that looks like this.

The damped harmonic oscillator (38)!

This is exactly the situation of Figure 38. We end up with a number for ϕ_k for each $\vec{k}$. Waves stretch, and eventually come to rest at some non-zero point.

That would be the end of the story, if it were not for quantum mechanics.

Quantum mechanics has zero-point oscillations. These can be thought of as classical waves. In space there is an oscillating field for every $\vec{k}$. Typically we don't see this because the frequency is so large. But in Figure 38 it comes to rest with the field just frozen. It is important to know when we reach the critical damping $\gamma = 2\omega$: the time will depend on $\vec{k}$: the smaller the wavelength, the later the wave will get frozen.

$$\begin{aligned} \gamma =& 2\omega \\ 3H =& 2\frac{k}{a} \end{aligned}$$

$$a = \frac{2k}{3H} \text{ is when the transition happens}$$

This transition is called "exiting the horizon".

For large k, rapid oscillation, freezing doesn't occur until we get a large a. Let us write in terms of wavelength.

$$\begin{aligned} k =& \frac{2\pi a(t)}{\lambda}, \text{ so crossover happens when} \\ a \approx& \frac{1}{\lambda H} \\ \lambda \approx& \frac{1}{H} \end{aligned}$$

λ is the distance where expansion matches speed of light–the Hubble distance or Hubble horizon. During inflation this distance was small, but it is the distance at which wave stops oscillating.

Imagine if new waves arise as existing waves reach horizon. Where do they come from? The vacuum fluctuations. These waves ruin the isotropy.

All this takes place on top of something else that is going on. In Figure 39 the field is gradually sliding down the potential, classically, plus the quantum noise. The latter has $\hbar$s in front, so it is tiny. It isn't enough to explain the energy isotropy.

What happens when we slide over and go down the edge in Figure 39? Ignore the noise, and assume the whole universe goes over: the potential turns into particles and heat; the expansion slows. It is no longer exponential, and the energy density dilutes.

Not quite. The noise says that the field is not quite the same at every point: we have a composite noise from the waves that got frozen. We get a complicated mixture as different places go over the cliff. By tweaking parameters, we get the observed $\frac{\delta\rho}{\rho}$

CMB comes from 200,000 years after falling off the cliff.

LS doesn't think NS variation significant.

Glossary

N_P Number of protons.

N_q Number of quarks.

$N_{\bar{P}}$ Number of antiprotons.

$N_{\bar{q}}$ Number of antiquarks.

N_γ Number of photons.

N_{e^+} Number of positrons.

N_{e^-} Number of electrons.

Λ Cosmological Constant.

κ Indicates curvature: {0,+1,-1}.

λ wavelength.

ν Frequency.

k_B Boltzmann's Constant.

Baryon number $\frac{1}{3}(N_q - N_{\bar{q}})$.

CMB Cosmic Microwave Background.

e-folding The time interval in which an exponentially growing quantity increases by a factor of e; it is the base-e analog of doubling time.

H Hubble's constant.

homogenous Homogeneity means the space is everywhere the same. Somebody located at a particular place in the space looks around them, see everywhere around them, and sees exactly the same thing that somebody else sees in any other position[1, Cosmology, Lecture 3].

isotropic Isotropy is uniformity in all orientations.

peculiar velocity The components of a galaxy's velocity that deviate from the Hubble flow.

scale factor The expansion of the universe is parameterized by a dimensionless scale factor a–Equation (1).

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